

Read these two documents fully from beginning to end. Then please focus on Chapter 3. I do not understand the derivations of equations in Chapter 3 at all, especially for the lateral load computation and the concept of no-roll-point/axis. Can you show me the important derivations of equations in Chapter 3 as detailed as possible step-by-step and explain to me what is the no-roll-point/axis and how does that affect the derivations?

已思考 2m 28s

I will use the book's Chapter 3 notation, but with the errata applied. The important errata for your question are: the vertical aerodynamic loads must distinguish Z_1^a and Z_2^a , and the printed Eq. (3.110) suspension-roll expressions need correction. The errata also corrects some tire-slip and friction-limit formulas later in Chapter 3.

978-94-017-8533-4SVDerrata

1. The big idea of Chapter 3

Chapter 3 is built like this:

kinematics $\rightarrow$ tire slips $\rightarrow$ tire forces $\rightarrow$ vehicle equilibrium $\rightarrow$ vertical loads

The confusing part is that global vehicle equilibrium gives only the total lateral load-transfer moment, not the front/rear split. That is why the book then studies the suspension geometry and introduces the no-roll centers / no-roll axis.

The key result is:

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = Y h$$

but this one equation cannot determine both ΔZ_1 and ΔZ_2 . The split depends on suspension geometry, tire vertical stiffness, and suspension roll stiffness.

2. Basic planar vehicle kinematics

The vehicle body is treated as a rigid body moving on a flat road. At the center of mass G ,

$$V_G = u \mathbf{i} + v \mathbf{j}$$

where u is forward speed and v is lateral speed in the vehicle frame. The yaw rate is

$$\Omega = r \mathbf{k}$$

Because the vehicle frame rotates with yaw rate r , the acceleration of G is not simply $\dot{u} \mathbf{i} + \dot{v} \mathbf{j}$. Differentiate V_G :

$$a_G = \frac{d}{dt}(u \mathbf{i} + v \mathbf{j}) = \dot{u} \mathbf{i} + \dot{u} \mathbf{j} + \dot{v} \mathbf{j} + v \dot{\mathbf{j}}$$

Since

$$\dot{\mathbf{i}} = r \mathbf{j}, \quad \dot{\mathbf{j}} = -r \mathbf{i}$$

we get

$$a_G = (\dot{u} - vr) \mathbf{i} + (\dot{v} + ur) \mathbf{j}$$

So Chapter 3 defines

$$a_x = \dot{u} - vr$$

$$a_y = \dot{v} + ur$$

This is one of the first major ideas: a_y is not $\ddot{y}$. It is the lateral acceleration component in the rotating vehicle frame.

3. Global in-plane equilibrium

The tire forces are first transformed into vehicle-frame forces. For each wheel,

$$X_{ij} = F_{xij} \cos \delta_{ij} - F_{yij} \sin \delta_{ij}$$

$$Y_{ij} = F_{xij} \sin \delta_{ij} + F_{yij} \cos \delta_{ij}$$

Then define axle sums:

$$X_1 = X_{11} + X_{12}, \quad X_2 = X_{21} + X_{22}$$

$$Y_1 = Y_{11} + Y_{12}, \quad Y_2 = Y_{21} + Y_{22}$$

and left/right half-differences:

$$\Delta X_1 = \frac{X_{12} - X_{11}}{2}, \quad \Delta X_2 = \frac{X_{22} - X_{21}}{2}$$

The planar dynamic equations become

$$m(\dot{u} - vr) = X_1 + X_2 - X_a$$

$$m(\dot{v} + ur) = Y_1 + Y_2$$

$$J_s \dot{r} = Y_1 a_1 - Y_2 a_2 + \Delta X_1 t_1 + \Delta X_2 t_2$$

Define

$$Y = Y_1 + Y_2 = m a_y$$

$$N_Y = Y_1 a_1 - Y_2 a_2$$

$$N_X = \Delta X_1 t_1 + \Delta X_2 t_2$$

so that

$$J_s \dot{r} = N_Y + N_X$$

Therefore

$$N_Y = J_s \dot{r} - N_X$$

Now solve

$$Y = Y_1 + Y_2$$

$$N_Y = Y_1 a_1 - Y_2 a_2$$

for Y_1, Y_2 . Since $l = a_1 + a_2$,

$$Y_1 = \frac{Y a_2 + N_Y}{l}$$

$$Y_2 = \frac{Y a_1 - N_Y}{l}$$

This says: the front/rear lateral-force split is determined by global force and yaw-moment equilibrium. It is not determined by the suspension.

4. Vertical load definitions

For each axle,

$$Z_1 = Z_{11} + Z_{12}$$

$$Z_2 = Z_{21} + Z_{22}$$

The lateral load transfers are defined as

$$\Delta Z_1 = \frac{Z_{12} - Z_{11}}{2}$$

$$\Delta Z_2 = \frac{Z_{22} - Z_{21}}{2}$$

Therefore,

$$Z_{11} = \frac{Z_1}{2} - \Delta Z_1, \quad Z_{12} = \frac{Z_1}{2} + \Delta Z_1$$

$$Z_{21} = \frac{Z_2}{2} - \Delta Z_2, \quad Z_{22} = \frac{Z_2}{2} + \Delta Z_2$$

So once you know $Z_1, Z_2, \Delta Z_1, \Delta Z_2$, you know all four tire vertical loads.

5. Longitudinal load transfer

The vertical force balance is, with errata applied,

$$Z_1 + Z_1^a + Z_2 + Z_2^a = mg$$

The pitch moment balance gives approximately

$$(Z_1 + Z_1^a) a_1 - (Z_2 + Z_2^a) a_2 = -m a_x h$$

ignoring the small $J_{xx} r^2$ term.

Solving gives

$$Z_1 = Z_1^0 + \Delta Z - Z_1^a$$

$$Z_2 = Z_2^0 - \Delta Z - Z_2^a$$

where

$$Z_1^0 = \frac{mg a_2}{l}$$

$$Z_2^0 = \frac{mg a_1}{l}$$

and

$$\Delta Z \approx -\frac{m a_x h}{l}$$

Interpretation:

When braking, $a_x < 0$, so

$$\Delta Z > 0$$

and the front axle load Z_1 increases. When accelerating, $a_x > 0$, so

$$\Delta Z < 0$$

and the rear axle load Z_2 increases.

6. The lateral load-transfer equation

The roll moment balance gives

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = Yh + J_{xx} \ddot{\phi}$$

Usually $J_{xx} \ddot{\phi}$ is small, so

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 \approx Yh = ma_y h$$

This is the most important lateral load-transfer equation.

But it is only **one equation** for two unknowns:

$$\Delta Z_1, \quad \Delta Z_2$$

So global equilibrium tells you the **total lateral load-transfer moment**, but not the front/rear distribution.

That is why Chapter 3 needs the suspension analysis.

7. Why the no-roll center is introduced

For each axle, the suspension linkage has an instantaneous geometric property. In the simplified first-order suspension analysis, you can construct a point Q_i with height q_i , such that:

A lateral force applied at Q_i is transmitted through the suspension links without producing suspension spring roll.

That point is traditionally called the roll center. Guiggiani argues it is better called the **no-roll center**, because the defining property is not "the body rolls about this point." The defining property is:

$$\text{force through } Q_i \quad \Rightarrow \quad \text{no suspension-roll spring deflection}$$

For a two-axle vehicle there are two such points:

$$Q_1 \quad \text{front no-roll center}$$

$$Q_2 \quad \text{rear no-roll center}$$

The line connecting them is traditionally called the roll axis, but the book's preferred interpretation is:

$$\text{no-roll axis} = \text{line through the two no-roll centers}$$

A force whose line of action passes through this axis produces no **suspension** roll. But this does **not** mean no tire deflection, no jacking, and no load transfer.

8. Decomposing the lateral force using the no-roll axis

Suppose the vehicle has lateral force requirement

$$Y = Y_1 + Y_2$$

acting at height h . The front and rear lateral forces Y_1, Y_2 are already fixed by global equilibrium:

$$Y_1 = \frac{Y a_2 + N_Y}{l}$$

$$Y_2 = \frac{Y a_1 - N_Y}{l}$$

Now place an equivalent point Q on the no-roll axis. Its height is q^h . Since Q is the point on the line between Q_1 and Q_2 corresponding to the force split Y_1, Y_2 ,

$$q^h = \frac{Y_1 q_1 + Y_2 q_2}{Y}$$

Substitute the formulas for Y_1, Y_2 :

$$q^h = \frac{a_2 q_1 + a_1 q_2}{l} + \frac{N_Y}{Y} \frac{q_1 - q_2}{l}$$

Equivalently,

$$q^h = \frac{a_2 q_1 + a_1 q_2}{l} - \frac{N_Y}{Y} \frac{q_2 - q_1}{l}$$

That sign is important; it is one of the places where you should be careful with the errata.

Now the lateral force at height h can be decomposed into:

1. a lateral force Y_1 applied at the front no-roll center Q_1 ;
2. a lateral force Y_2 applied at the rear no-roll center Q_2 ;
3. a remaining roll moment

$$Y(h - q^h)$$

So the actual lateral force at height h is equivalent to

$$Y_1 \text{ through } Q_1 + Y_2 \text{ through } Q_2 + Y(h - q^h)$$

The book emphasizes that the lever arm is the **vertical distance**

$$h - q^h$$

not the shortest distance to the no-roll axis.

9. The two pieces of lateral load transfer

The lateral load transfer at each axle has two contributions.

9.1 Geometric / no-roll-center contribution

Because Y_i acts through Q_i , at height q_i , the suspension linkage transmits this force to the ground with a vertical left/right load difference:

$$\Delta Z_i^Y t_i = Y_i q_i$$

So

$$\Delta Z_i^Y = \frac{Y_i q_i}{t_i}$$

This part happens even without suspension spring roll.

If q_i is high, the direct geometric contribution is large. If $q_i = 0$, this direct contribution vanishes.

9.2 Elastic roll contribution

The remaining roll moment is

$$Y(h - q^h)$$

This is resisted by suspension roll stiffness:

$$Y(h - q^h) = k_{\phi 1}^s \phi_1^s + k_{\phi 2}^s \phi_2^s$$

The corresponding elastic load-transfer part is

$$\Delta Z_i^E t_i = k_{\phi i}^s \phi_i^s$$

Therefore the total axle lateral load transfer is

$$\Delta Z_i t_i = Y_i q_i + k_{\phi i}^s \phi_i^s$$

The same load transfer also appears through tire vertical deflection:

$$\Delta Z_i t_i = k_{\phi i}^p \phi_i^p$$

So the book writes the fundamental internal-equilibrium relation as

$$\Delta Z_i t_i = Y_i q_i + k_{\phi i}^s \phi_i^s = k_{\phi i}^p \phi_i^p$$

This is the core of the lateral load-transfer derivation.

10. Why tire vertical stiffness appears

The total vehicle roll angle at each axle is split into suspension roll and tire-deflection roll:

$$\phi = \phi_i^s + \phi_i^p$$

The tire roll stiffness is

$$k_{\phi i}^p = \frac{p_i t_i^2}{2}$$

where p_i is the vertical stiffness of one tire on axle i .

The suspension and tire act like springs in series for roll:

$$k_{\phi i} = \frac{k_{\phi i}^s k_{\phi i}^p}{k_{\phi i}^s + k_{\phi i}^p}$$

The total roll stiffness is

$$k_{\phi} = k_{\phi 1} + k_{\phi 2}$$

This is why the tire vertical stiffness cannot be ignored in a serious race-car load-transfer analysis. The tire is not simply a rigid vertical support.

11. Solving the full first-order suspension system

The full system is:

$$\phi = \phi_1^s + \phi_1^p = \phi_2^s + \phi_2^p$$

$$Y(h - q^h) = k_{\phi 1}^s \phi_1^s + k_{\phi 2}^s \phi_2^s$$

$$Y_1 q_1 + k_{\phi 1}^s \phi_1^s = k_{\phi 1}^p \phi_1^p$$

$$Y_2 q_2 + k_{\phi 2}^s \phi_2^s = k_{\phi 2}^p \phi_2^p$$

Solving this linear system gives the explicit lateral load transfers:

$$\Delta Z_1 = \frac{1}{t_1} \left[k_{\phi 1} Y(h - q^h) + Y_1 q_1 + \frac{k_{\phi 1} k_{\phi 2}}{k_{\phi}} \left(\frac{Y_2 q_2}{k_{\phi 2}^p} - \frac{Y_1 q_1}{k_{\phi 1}^p} \right) \right]$$

$$\Delta Z_2 = \frac{1}{t_2} \left[k_{\phi 2} Y(h - q^h) + Y_2 q_2 + \frac{k_{\phi 1} k_{\phi 2}}{k_{\phi}} \left(\frac{Y_1 q_1}{k_{\phi 1}^p} - \frac{Y_2 q_2}{k_{\phi 2}^p} \right) \right]$$

These are the important full Chapter 3 lateral load-transfer formulas with tire vertical compliance included.

The terms have clear meanings:

$$\frac{k_{\phi i}}{k_{\phi}} Y(h - q^h)$$

is the elastic roll-stiffness distribution term.

$$Y_i q_i$$

is the direct geometric / no-roll-center term.

The final bracket involving $k_{\phi i}^p$ is the correction caused by tire vertical compliance.

12. Rigid-tire simplification

If tire vertical compliance is neglected,

$$k_{\phi i}^p \rightarrow \infty$$

so

$$k_{\phi i} \rightarrow k_{\phi i}^s$$

and the formulas become much simpler:

$$k_{\psi}\phi = Y(h-q^b)$$

$$\Delta Z_i t_i = \frac{k_{\phi i}}{k_{\phi}} Y(h-q^b) + Y_i q_i$$

or equivalently,

$$\Delta Z_i t_i = k_{\phi i} \phi + Y_i q_i$$

This is probably the cleanest formula for intuition.

It says:

lateral load transfer = elastic roll stiffness contribution + geometric no-roll-center contribution

13. Final tire vertical loads

After computing $\Delta Z_1, \Delta Z_2$, the four tire loads are:

$$Z_{11} = \frac{1}{2} \left(Z_1^0 - Z_1^a + \Delta Z \right) - \Delta Z_1$$

$$Z_{12} = \frac{1}{2} \left(Z_1^0 - Z_1^a + \Delta Z \right) + \Delta Z_1$$

$$Z_{21} = \frac{1}{2} \left(Z_2^0 - Z_2^a - \Delta Z \right) - \Delta Z_2$$

$$Z_{22} = \frac{1}{2} \left(Z_2^0 - Z_2^a - \Delta Z \right) + \Delta Z_2$$

where

$$Z_1^0 = \frac{mga_2}{l}, \quad Z_2^0 = \frac{mga_1}{l}$$

$$\Delta Z \approx -\frac{ma_x h}{l}$$

and $\Delta Z_1, \Delta Z_2$ come from the suspension/no-roll-center derivation above.

14. What the no-roll axis really means

The no-roll axis is not a physical hinge axis of the vehicle body.

It is not saying:

the chassis literally rolls about this line

The correct interpretation is:

A force through the no-roll axis produces no suspension spring roll.

But it may still produce:

tire deflection

lateral load transfer

suspension jacking

vertical displacement

This is why Guiggiani criticizes the classical phrase "the vehicle rolls about the roll axis." A better mental model is:

no-roll axis = a force-decomposition tool for suspension load transfer

not

a rigid-body rotation axis

The position of the no-roll centers q_1, q_2 affects the derivation because it decides how much of Yh is handled geometrically and how much remains as elastic roll moment:

$$Yh = \underbrace{Yq^b}_{\text{geometric part}} + \underbrace{Y(h-q^b)}_{\text{elastic roll part}}$$

Higher q^b reduces the elastic roll moment $Y(h-q^b)$, but it does not eliminate load transfer. It merely changes the path by which the load transfer is generated.

That is the central idea of Chapter 3's lateral load computation.

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Please double check your above explanations of Chapter 3. Make sure that not only the explanations are fully correct, but also that the important equations are derived step-by-step correctly and explained correctly especially for the no roll axis concept and how it affects load transfer. Because Chapter 3 is the basic of Chapter 9, I don't think I can fully understand Chapter 9 if I cannot understand Chapter 3. So please be thorough on Chapter 3 and help me understand.

已思考 2m 22s ›

I rechecked the Chapter 3 derivation chain and the errata. My previous explanation was directionally correct, but the most important refinements are these:

1. The no-roll axis does not mean the car body literally rolls about that axis. It means a force whose line of action passes through that axis produces no suspension-spring roll, although it may still produce tire deflection, load transfer, jacking, and vertical displacement.
2. The corrected formula for q^b must follow the errata sign convention.
3. The corrected Eq. (3.110) from the errata is important; the printed book formula has errors.
4. The cleanest full lateral-load-transfer formula is Eq. (3.114), not the more expanded Eq. (3.113).
5. In Chapter 3, roll is treated quasi-statically as an internal suspension/tire coordinate. Roll inertia is neglected. Chapter 9 later adds true roll motion dynamics.

The book itself says that Chapter 3's model is built from congruence/kinematic equations, equilibrium equations, and constitutive tire equations, and then it separately analyzes the suspensions to compute vertical loads. 978-94-017-8533-4

1. Start from the planar vehicle model

Chapter 3 assumes the vehicle body moves on a flat road with body-fixed axes:

$\mathbf{i}$ = forward, $\mathbf{j}$ = lateral, $\mathbf{k}$ = vertical upward.

The velocity of the center of mass G is

$$V_G = u\mathbf{i} + v\mathbf{j}$$

and the yaw angular velocity is

$$\Omega = r\mathbf{k}.$$

Because $\mathbf{i}, \mathbf{j}$ rotate with the car,

$$\dot{\mathbf{i}} = r\mathbf{j}, \quad \dot{\mathbf{j}} = -r\mathbf{i}.$$

Therefore,

$$a_G = \frac{d}{dt}(u\mathbf{i} + v\mathbf{j}) = \dot{u}\mathbf{i} + ur\mathbf{j} + \dot{v}\mathbf{j} - vr\mathbf{i}.$$

So

$$a_G = (\dot{u} - vr)\mathbf{i} + (\dot{v} + ur)\mathbf{j}$$

and Chapter 3 defines

$$a_x = \dot{u} - vr$$

$$a_y = \dot{v} + ur$$

This matters because later

$$Y = ma_y$$

is the required total lateral road force.

2. Road-tire forces in the vehicle frame

Each tire has its own wheel-frame longitudinal and lateral forces:

$$F_{xij}, \quad F_{yij}.$$

But the vehicle equilibrium needs forces in the vehicle frame. If wheel ij has steer angle δ_{ij} , then

$$X_{ij} = F_{xij} \cos \delta_{ij} - F_{yij} \sin \delta_{ij}$$

$$Y_{ij} = F_{xij} \sin \delta_{ij} + F_{yij} \cos \delta_{ij}.$$

Then define axle sums:

$$X_1 = X_{11} + X_{12}, \quad X_2 = X_{21} + X_{22}$$

$$Y_1 = Y_{11} + Y_{12}, \quad Y_2 = Y_{21} + Y_{22}$$

and left-right half-differences:

$$\Delta X_1 = \frac{X_{12} - X_{11}}{2}, \quad \Delta X_2 = \frac{X_{22} - X_{21}}{2}$$

$$\Delta Z_1 = \frac{Z_{12} - Z_{11}}{2}, \quad \Delta Z_2 = \frac{Z_{22} - Z_{21}}{2}.$$

The book defines these tire-force transformations and vertical-load half-differences before deriving the explicit equilibrium equations. 978-94-017-8533-4

3. Global equilibrium gives Y_1, Y_2 , but not $\Delta Z_1, \Delta Z_2$

The three planar dynamic equations are

$$ma_x = X_1 + X_2 - X_a$$

$$ma_y = Y_1 + Y_2$$

$$J_z \dot{r} = Y_1 a_1 - Y_2 a_2 + \Delta X_1 t_1 + \Delta X_2 t_2.$$

Define

$$Y = Y_1 + Y_2 = ma_y$$

and

$$N_X = \Delta X_1 t_1 + \Delta X_2 t_2$$

$$N_Y = Y_1 a_1 - Y_2 a_2.$$

Then

$$J_z \dot{r} = N_Y + N_X$$

so

$$N_Y = J_z \dot{r} - N_X$$

Now solve

$$Y_1 + Y_2 = Y$$

$$Y_1 a_1 - Y_2 a_2 = N_Y.$$

From the first equation,

$$Y_2 = Y - Y_1.$$

Substitute:

$$Y_1 a_1 - (Y - Y_1) a_2 = N_Y$$

$$Y_1 a_1 - Y a_2 + Y_1 a_2 = N_Y$$

$$Y_1 (a_1 + a_2) = Y a_2 + N_Y.$$

Since

$$l = a_1 + a_2,$$

we get

$$Y_1 = \frac{Y a_2 + N_Y}{l}$$

and

$$Y_2 = \frac{Y a_1 - N_Y}{l}$$

This is important: for a given vehicle motion and yaw acceleration, global equilibrium fixes the required front and rear lateral forces. Suspension geometry does not choose Y_1, Y_2 . The suspension decides how the corresponding vertical load transfers are distributed. Chapter 3 explicitly says Y_1, Y_2 come from global equilibrium and are not affected by suspension, tire type, or grip level in this algebraic equilibrium sense.

978-94-017-8533-4

4. Longitudinal load transfer

The vertical axle loads are

$$Z_1 = Z_{11} + Z_{12}, \qquad Z_2 = Z_{21} + Z_{22}.$$

The vertical force balance, with the errata correction, is

$$Z_1 + Z_1^0 + Z_2 + Z_2^0 = mg.$$

The pitch moment balance is

$$(Z_1 + Z_1^0) a_1 - (Z_2 + Z_2^0) a_2 = -ma_x h + J_{xz} r^2.$$

Let

$$A = Z_1 + Z_1^0, \qquad B = Z_2 + Z_2^0.$$

Then

$$A + B = mg$$

$$Aa_1 - Ba_2 = -ma_x h + J_{xz} r^2.$$

Solving gives

$$A = \frac{mga_2 - ma_x h + J_{xz} r^2}{l}$$

$$B = \frac{mga_1 + ma_x h - J_{xz} r^2}{l}.$$

Define

$$Z_1^0 = \frac{mga_2}{l}, \qquad Z_2^0 = \frac{mga_1}{l}$$

and

$$\Delta Z = -\frac{ma_x h}{l} + \frac{J_{xz} r^2}{l} \approx -\frac{ma_x h}{l}$$

so

$$Z_1 = Z_1^0 + \Delta Z - Z_1^a$$

$$Z_2 = Z_2^0 - \Delta Z - Z_2^a$$

This matches Chapter 3's longitudinal-load-transfer equations; the errata also corrects the duplicated aerodynamic-load subscript in the book's equilibrium equations.

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SV06errata

Interpretation:

- braking: $a_x < 0$, so $\Delta Z > 0$, front axle load increases;
- acceleration: $a_x > 0$, so $\Delta Z < 0$, rear axle load increases.

5. Global lateral load transfer

The roll moment equilibrium from Chapter 3 gives

$$-J_{zx} \dot{r} = Yh - \Delta Z_1 t_1 - \Delta Z_2 t_2.$$

Therefore,

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = Yh + J_{zx} \dot{r}$$

or, since $J_{zx} \dot{r}$ is usually small,

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 \approx Yh = ma_y h$$

This equation is only one scalar equation for two unknowns:

$$\Delta Z_1, \qquad \Delta Z_2.$$

So global equilibrium determines the total lateral load-transfer moment, but not its front/rear split. Chapter 3 explicitly states that one equation is not enough to obtain $\Delta Z_1, \Delta Z_2$, and that suspension geometry and compliances directly influence $\Delta Z_1/\Delta Z_2$.

978-94-017-8533-4

This is the central reason Chapter 3 needs the no-roll centers.

6. What the no-roll center really is

For each axle, Chapter 3 constructs a point Q_i as follows:

- find the instantaneous center B_i of each wheel hub relative to the body;
- take the contact patch center A_i ;
- draw the lines $A_i B_i$ from both left and right sides;
- their intersection gives Q_i .

The signed height of Q_i above the road is

$$q_i.$$

A point below the road has

$$q_i < 0.$$

For a two-axle vehicle, there are two such points:

$$Q_1, \qquad Q_2.$$

The line through them is traditionally called the roll axis, but Chapter 3 argues that the better name is **no-roll axis**.

978-94-017-8533-4

The defining property is:

A force applied through Q_i produces no suspension-spring roll at axle i .

That does **not** mean no load transfer.

That does **not** mean no tire deflection.

That does **not** mean the car body physically rotates about Q_i .

It only means:

the suspension springs do not need to deflect in roll to transmit that force.

Chapter 3 says a force through the roll axis may still affect vehicle roll angle because of tire deflections, but with no contribution from suspension deflections; it may also change the vertical displacements z_1, z_2 .

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This is the conceptual correction you need before Chapter 9.

7. Why the no-roll axis affects load transfer

Suppose the vehicle requires total lateral force

$$Y = ma_y,$$

In Chapter 3's suspension analysis, this is represented as a body force

$$-Y \mathbf{j}$$

acting at some point P , height h above the road.

From global equilibrium, the corresponding axle lateral forces must be

$$Y_1 = \frac{Y a_2^b}{l}, \quad Y_2 = \frac{Y a_1^b}{l}.$$

Here a_1^b, a_2^b are the effective longitudinal distances from the applied lateral force to the front and rear axes. They differ from a_1, a_2 when $N_Y \neq 0$.

Now choose a point Q on the no-roll axis directly below P . Its height is obtained by linear interpolation between Q_1 and Q_2 :

$$q^b = \frac{a_2^b q_1 + a_1^b q_2}{a_1^b + a_2^b}$$

Since

$$\begin{aligned} a_1^b + a_2^b &= l, \\ q^b &= \frac{a_2^b q_1 + a_1^b q_2}{l}, \end{aligned}$$

Equivalently, because

$$Y_1 = \frac{Y a_2^b}{l}, \quad Y_2 = \frac{Y a_1^b}{l},$$

we get

$$Y q^b = Y_1 q_1 + Y_2 q_2$$

or

$$q^b = \frac{Y_1 q_1 + Y_2 q_2}{Y}$$

Using

$$a_1^b = a_1 - \frac{N_Y}{Y}, \quad a_2^b = a_2 + \frac{N_Y}{Y},$$

we obtain

$$q^b = \frac{a_2 q_1 + a_1 q_2}{l} + \frac{N_Y}{Y} \frac{q_1 - q_2}{l}.$$

Equivalently,

$$q^b = \frac{a_2 q_1 + a_1 q_2}{l} - \frac{N_Y}{Y} \frac{q_2 - q_1}{l}$$

This sign is important. The errata says the sign in front of the N_Y/Y term in Eq. (3.101') must be changed. SVDeirasa

8. Decomposing the lateral force

The lateral force $-Y\mathbf{j}$ applied at P , height h , is equivalent to:

- the same lateral force $-Y\mathbf{j}$ applied at Q , height q^b ;
- plus a pure roll moment

$$Y(h - q^b)$$

Then the force at Q can be decomposed into:

$$-Y_1\mathbf{j} \quad \text{at } Q_1$$

and

$$-Y_2\mathbf{j} \quad \text{at } Q_2.$$

Thus the full equivalence is:

$$-Y\mathbf{j} \text{ at } P \equiv -Y_1\mathbf{j} \text{ at } Q_1 + -Y_2\mathbf{j} \text{ at } Q_2 + Y(h - q^b)\mathbf{i}$$

Chapter 3 emphasizes that the moment arm is the **vertical distance**

$$h - q^b,$$

not the shortest distance from P to the no-roll axis. 978-94-017-8533-4

This is a key conceptual point.

9. The two parts of lateral load transfer

At each axle,

$$\Delta Z_i t_i$$

is the roll moment produced by the left-right vertical load difference.

Chapter 3 splits it into two pieces:

$$\Delta Z_i t_i = \underbrace{Y_i q_i}_{\text{geometric / no-roll-center part}} + \underbrace{k_{\phi i}^s \phi_i^s}_{\text{suspension elastic part}}$$

and because the same vertical load difference also deforms the tires,

$$\Delta Z_i t_i = Y_i q_i + k_{\phi i}^s \phi_i^s = k_{\phi i}^p \phi_i^p$$

This is Eq. (3.104) in meaning. Chapter 3 states that the load transfer is the sum of the part due to the suspension linkage and the part due to the suspension springs. 978-94-017-8533-4

The first term,

$$Y_i q_i,$$

is the **geometric load transfer**.

It exists even when there is no suspension roll.

The second term,

$$k_{\phi i}^s \phi_i^s,$$

comes from suspension spring/anti-roll-bar deflection.

So the no-roll center changes the load-transfer derivation because it decides how much of $\mathcal{Y}\hbar$ goes directly through suspension geometry and how much remains as elastic roll moment.

10. Why force through the no-roll center still causes load transfer

This is probably the easiest place to misunderstand Chapter 3.

Suppose a lateral force Y_i acts at Q_i , height q_i .

Because Q_i is a no-roll center, the suspension links can transmit that force without suspension spring roll.

But the force is still applied at height q_i . To balance its moment about the ground plane, the road must create a left-right vertical force difference:

$$\Delta Z_i^Y t_i = Y_i q_i$$

Therefore:

$$\text{no suspension roll} \neq \text{no load transfer}$$

Chapter 3 explicitly gives this load transfer without suspension roll and then discusses suspension jacking. 978-94-017-8533-4

So the correct mental picture is:

force through no-roll center $\Rightarrow$ load transfer through geometry $\Rightarrow$ possibly tire deflection/jacking $\Rightarrow$ but no suspension spring roll

not:

$$\text{force through no-roll center} \Rightarrow \text{nothing happens.}$$

11. Roll stiffnesses: suspension and tire are in series

For axle i , Chapter 3 defines:

$$k_{\phi i}^s = \text{suspension roll stiffness}$$

$$k_{\phi i}^p = \text{tire roll stiffness}$$

$$k_{\phi i} = \text{combined axle roll stiffness.}$$

The tire roll stiffness is

$$k_{\phi i}^p = \frac{p_i t_i^2}{2}$$

where p_i is the vertical stiffness of one tire on axle i .

Because tire and suspension roll compliances are in series,

$$k_{\phi i} = \frac{k_{\phi i}^s k_{\phi i}^p}{k_{\phi i}^s + k_{\phi i}^p}$$

and the total vehicle roll stiffness is

$$k_{\phi} = k_{\phi 1} + k_{\phi 2}$$

Chapter 3 derives these stiffness definitions by applying a pure rolling moment and measuring the resulting load transfer; it also notes that tire vertical compliance should not be ignored, especially for formula cars. 978-94-017-8533-4

12. Solving the full lateral-load-transfer system

Now define, for compactness,

$$U_1 = Y_1 q_1, \quad U_2 = Y_2 q_2,$$

$$M_s = Y(h - q^b).$$

Here M_s is the moment that must be handled by suspension roll stiffness.

Let

$$A_1 = k_{\phi 1}^s, \quad A_2 = k_{\phi 2}^s$$

$$P_1 = k_{\phi 1}^p, \quad P_2 = k_{\phi 2}^p$$

$$K_1 = k_{\phi 1}, \quad K_2 = k_{\phi 2}, \quad K = K_1 + K_2.$$

The unknown roll pieces satisfy:

$$\begin{aligned}\phi &= \phi_1^s + \phi_1^p \\ \phi &= \phi_2^s + \phi_2^p \\ M_s &= A_1 \phi_1^s + A_2 \phi_2^s \\ U_1 + A_1 \phi_1^s &= P_1 \phi_1^p \\ U_2 + A_2 \phi_2^s &= P_2 \phi_2^p.\end{aligned}$$

The last two equations say:

$$\text{geometric moment} + \text{suspension spring moment} = \text{tire-deflection moment}.$$

Now use

$$\phi_1^p = \phi - \phi_1^s.$$

For axle 1:

$$\begin{aligned}U_1 + A_1 \phi_1^s &= P_1 (\phi - \phi_1^s) \\ U_1 + A_1 \phi_1^s &= P_1 \phi - P_1 \phi_1^s \\ (A_1 + P_1) \phi_1^s &= P_1 \phi - U_1 \\ \phi_1^s &= \frac{P_1 \phi - U_1}{A_1 + P_1}\end{aligned}$$

Similarly,

$$\phi_2^s = \frac{P_2 \phi - U_2}{A_2 + P_2}$$

Substitute these into

$$M_s = A_1 \phi_1^s + A_2 \phi_2^s.$$

Then

$$M_s = A_1 \frac{P_1 \phi - U_1}{A_1 + P_1} + A_2 \frac{P_2 \phi - U_2}{A_2 + P_2}.$$

Since

$$K_1 = \frac{A_1 P_1}{A_1 + P_1}, \quad K_2 = \frac{A_2 P_2}{A_2 + P_2},$$

and

$$\frac{A_1}{A_1 + P_1} = \frac{K_1}{P_1}, \quad \frac{A_2}{A_2 + P_2} = \frac{K_2}{P_2},$$

we get

$$M_s = K_1 \phi - \frac{K_1}{P_1} U_1 + K_2 \phi - \frac{K_2}{P_2} U_2.$$

Therefore,

$$M_s = K \phi - \frac{K_1}{P_1} U_1 - \frac{K_2}{P_2} U_2.$$

So

$$K \phi = M_s + \frac{K_1}{P_1} U_1 + \frac{K_2}{P_2} U_2$$

Returning to Chapter 3 notation:

$$k_\phi \phi = Y(h - q^b) + \frac{k_{\phi 1}}{k_{\phi 1}^p} Y_1 q_1 + \frac{k_{\phi 2}}{k_{\phi 2}^p} Y_2 q_2$$

This is the meaning of Eq. (3.112). It shows that the total roll angle is affected by both the elastic roll moment and the geometric load-transfer terms through tire compliance. Chapter 3 also notes that if q^b is almost constant, N_Y has little effect on roll angle but can strongly affect lateral load transfer by redistributing Y_1, Y_2 .
978-94-017-8533-4

13. Corrected suspension-roll deflection formulas

Using the same notation, the corrected suspension-roll deflections are:

$$\begin{aligned}\phi_1^s &= \frac{1}{k_{\phi 1}^p} \frac{k_{\phi 1} k_{\phi 2}}{k_\phi} \left[\frac{Y(h - q^b)}{k_{\phi 2}} - \frac{Y_1 q_1}{k_{\phi 1}^p} + \frac{Y_2 q_2}{k_{\phi 2}^p} \right] \\ \phi_2^s &= \frac{1}{k_{\phi 2}^p} \frac{k_{\phi 1} k_{\phi 2}}{k_\phi} \left[\frac{Y(h - q^b)}{k_{\phi 1}} - \frac{Y_2 q_2}{k_{\phi 2}^p} + \frac{Y_1 q_1}{k_{\phi 1}^p} \right]\end{aligned}$$

These are the errata-corrected Eq. (3.110). The errata explicitly says the last two terms in both equations must be changed, and it gives the corrected expression.
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This correction matters because otherwise the interaction between front/rear no-roll-center terms and tire compliance is wrong.

14. Final lateral load-transfer formulas

Now use

$$\begin{aligned}\Delta Z_1 t_1 &= U_1 + A_1 \phi_1^s \\ \Delta Z_2 t_2 &= U_2 + A_2 \phi_2^s.\end{aligned}$$

Substitute the corrected ϕ_i^s . We obtain

$$\begin{aligned}\Delta Z_1 &= \frac{1}{t_1} \left[k_{\phi 1} Y(h - q^b) + Y_1 q_1 + \frac{k_{\phi 1} k_{\phi 2}}{k_\phi} \left(\frac{Y_2 q_2}{k_{\phi 2}^p} - \frac{Y_1 q_1}{k_{\phi 1}^p} \right) \right] \\ \Delta Z_2 &= \frac{1}{t_2} \left[k_{\phi 2} Y(h - q^b) + Y_2 q_2 + \frac{k_{\phi 1} k_{\phi 2}}{k_\phi} \left(\frac{Y_1 q_1}{k_{\phi 1}^p} - \frac{Y_2 q_2}{k_{\phi 2}^p} \right) \right]\end{aligned}$$

These are the clean Eq. (3.114) lateral-load-transfer formulas. Chapter 3 says these transfers are linear functions of Y and N_Y , equivalently of Y_1, Y_2 , and that the simple global equation $\Delta Z_1 t_1 + \Delta Z_2 t_2 = Y h$ must always hold.
978-94-017-8533-4

Check the sum:

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = Y(h - q^b) + Y_1 q_1 + Y_2 q_2.$$

But

$$Y_1 q_1 + Y_2 q_2 = Y q^b.$$

So

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = Y(h - q^b) + Y q^b = Y h.$$

So the detailed suspension result is consistent with the global roll-moment equation.

That consistency check is very important.

15. Rigid-tire limit

If tires are treated as vertically rigid,

$$k_{\phi i}^p \rightarrow \infty.$$

Then

$$k_{\phi i} \rightarrow k_{\phi i}^s$$

and the tire-compliance correction terms disappear.

The total roll angle becomes

$$k_\phi \phi = Y(h - q^b)$$

and the lateral load transfer becomes

$$\Delta Z_i t_i = \frac{k_{\phi i}}{k_\phi} Y(h - q^b) + Y_i q_i$$

or equivalently,

$$\Delta Z_i t_i = k_{\phi i} \phi + Y_i q_i.$$

Chapter 3 gives exactly this simplification but warns that, especially in formula cars, assuming vertically rigid tires may be unsafe; tire compliance should be included through the full formula rather than by simply softening the suspension stiffness.
978-94-017-8533-4

This warning is very important for your own vehicle model.

16. Final four tire vertical loads

After obtaining $Z_1, Z_2, \Delta Z_1, \Delta Z_2$, the four tire vertical loads are

$$\begin{aligned}Z_{11} &= \frac{Z_1}{2} - \Delta Z_1 \\ Z_{12} &= \frac{Z_1}{2} + \Delta Z_1 \\ Z_{21} &= \frac{Z_2}{2} - \Delta Z_2 \\ Z_{22} &= \frac{Z_2}{2} + \Delta Z_2\end{aligned}$$

with

$$\begin{aligned}Z_1 &= Z_1^0 + \Delta Z - Z_1^a \\ Z_2 &= Z_2^0 - \Delta Z - Z_2^a.\end{aligned}$$

The book defines these vertical loads from axle totals and lateral half-differences before deriving the load-transfer formulas.
978-94-017-8533-4

17. What the no-roll axis changes physically

The no-roll axis affects lateral load transfer in three ways.

First, it creates direct geometric load transfer:

$$Y_i q_i.$$

Higher q_i means more of the lateral force at that axle is transmitted geometrically through the suspension links.

Second, it changes the remaining suspension roll moment:

$$Y(h - q^b).$$

Higher q^b means smaller suspension-spring roll moment, hence smaller suspension roll angle.

Third, it changes the front/rear load-transfer split. Even if the total

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = Yh$$

is fixed, the individual values $\Delta Z_1, \Delta Z_2$ depend on

$$q_1, \quad q_2, \quad k_{\phi 1}^s, \quad k_{\phi 2}^s, \quad k_{\phi 1}^p, \quad k_{\phi 2}^p.$$

So the no-roll axis does **not** change the total lateral load-transfer moment required by physics. It changes the **path** and **distribution** of that load transfer.

The best compact interpretation is:

$$Yh = \underbrace{Y q^b}_{\text{geometric load-transfer moment}} + \underbrace{Y(h - q^b)}_{\text{suspension elastic roll moment}}$$

where

$$Y q^b = Y_1 q_1 + Y_2 q_2.$$

18. The most important conceptual correction

Do not think:

roll axis = axis around which the car body rotates.

Think instead:

no-roll axis = the locus through which a force can pass without creating suspension-spring roll.

A force through that axis can still create load transfer:

$$\Delta Z_1^Y t_1 = Y_i q_i.$$

It can still create tire deflection.

It can still create jacking.

It can still affect total vehicle roll angle through tire compliance.

This is why Guiggiani prefers **no-roll center/axis** over classical **roll center/axis**. The old phrase "roll center" tempts you to imagine a physical hinge. Chapter 3's derivation shows it is really a **force-decomposition** concept.

19. Why this matters for Chapter 9

Chapter 3 gives the quasi-static load-transfer foundation:

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = Yh$$

plus the suspension/no-roll-axis split formulas.

Chapter 9 then introduces real 3D vehicle orientation and roll motion. But Chapter 9 still needs the Chapter 3 insight:

global dynamics gives total vertical-load moments;

suspension geometry and compliance distribute them among the tires.

So for Chapter 9, the mental model should be:

3D rigid-body dynamics + suspension/tire compliance + no-roll-axis force decomposition = individual tire vertical loads.

Chapter 3 is not just a simplified version of Chapter 9. It is the conceptual basis for understanding how lateral forces, roll moments, no-roll centers, tire compliance, and vertical loads are coupled.

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Double check whether you have fully explained Chapter 3 fully and correctly in order to help me fully understand. If not, add more details. If so, tell me then how Chapter 3 is related to Chapter 9's equations of motion and tyre load equations. Explain as detailed as possible too.

已思考 1m 45s ›

After rechecking: my previous explanation was correct for the central lateral-load-transfer derivation, but it was **not complete enough** if your goal is to use Chapter 3 as the foundation for Chapter 9. The missing layer is this:

Chapter 3 is not only "derive lateral load transfer." It is a closed vehicle-model architecture:

kinematics → tire slips → tire forces → global equilibrium → suspension/tire load-transfer closure → individual tire loads.

Chapter 9 then keeps most of this architecture, but replaces the quasi-static roll treatment with actual roll/pitch dynamics.

I will give you the corrected and fuller picture.

A. What Chapter 3 is really doing

Chapter 3 assumes a flat road and a vehicle whose main motion is planar:

$$V_G = u\mathbf{i} + v\mathbf{j}, \quad \Omega = r\mathbf{k}.$$

Because $\mathbf{i}, \mathbf{j}$ rotate with yaw rate r ,

$$a_G = (\dot{u} - vr)\mathbf{i} + (\dot{v} + ur)\mathbf{j}.$$

So define

$$a_x = \dot{u} - vr, \quad a_y = \dot{v} + ur.$$

That is the first pillar. The lateral tire forces must satisfy

$$Y = Y_1 + Y_2 = ma_y.$$

Chapter 3 then uses yaw equilibrium:

$$J_z \dot{r} = Y_1 a_1 - Y_2 a_2 + \Delta X_1 t_1 + \Delta X_2 t_2.$$

Define

$$N_X = \Delta X_1 t_1 + \Delta X_2 t_2,$$

$$N_Y = Y_1 a_1 - Y_2 a_2.$$

Then

$$N_Y = J_z \dot{r} - N_X.$$

Solving

$$Y_1 + Y_2 = Y, \quad Y_1 a_1 - Y_2 a_2 = N_Y$$

gives

$$Y_1 = \frac{Y a_2 + N_Y}{l}$$

$$Y_2 = \frac{Y a_1 - N_Y}{l}$$

This is important: the **front/rear lateral-force split** is fixed by global force and yaw-moment equilibrium, not by the suspension.

The suspension affects the vertical load split, not the required Y_1, Y_2 .

The book explicitly emphasizes that the lateral load transfers $\Delta Z_1, \Delta Z_2$ cannot be found from global equilibrium alone, and that suspension geometry and compliance directly influence their ratio.

978-94-017-8533-4

B. Longitudinal load transfer in Chapter 3

The vertical axle loads are

$$Z_1 = Z_{11} + Z_{12}, \quad Z_2 = Z_{21} + Z_{22}.$$

With the book's sign convention and errata correction for aerodynamic vertical loads,

$$Z_1 = Z_1^0 + \Delta Z - Z_1^a,$$

$$Z_2 = Z_2^0 - \Delta Z - Z_2^a,$$

where

$$Z_1^0 = \frac{mg a_2}{l}, \quad Z_2^0 = \frac{mg a_1}{l},$$

and

$$\Delta Z = -\frac{ma_x h}{l} + \frac{J_{zz} r^2}{l} \approx -\frac{ma_x h}{l}.$$

So under braking $a_x < 0$, $\Delta Z > 0$, and the front axle gains load.

This part does **not** depend on no-roll centers. Longitudinal load transfer comes mainly from the longitudinal inertial force acting at height h . 978-94-017-8533-4

C. Global lateral load transfer in Chapter 3

The roll-moment balance gives

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = Yh + J_{xx} \dot{r}.$$

Usually $J_{xx} \dot{r}$ is small, so

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 \approx Yh = ma_y h.$$

This equation is absolutely fundamental, but it is only **one equation** for two unknowns:

$$\Delta Z_1, \quad \Delta Z_2.$$

Therefore:

global equilibrium fixes total lateral load transfer, but not its front/rear distribution.

That is why Chapter 3 introduces suspension geometry, no-roll centers, roll stiffness, and tire vertical stiffness. 978-94-017-8533-4

D. The no-roll center and no-roll axis

For each axle, Chapter 3 constructs a point Q_i , height q_i , from the suspension geometry. The line through the front and rear points Q_1, Q_2 is traditionally called the roll axis, but Guiggiani prefers **no-roll axis**.

The correct meaning is:

A lateral force through the no-roll center causes no suspension-spring roll at that axle.

It does **not** mean:

the car body literally rolls about this point or line.

It also does not mean:

no load transfer.

A force through the no-roll center can still cause:

geometric load transfer,

tire deflection,

suspension jacking,

body vertical displacement.

This is the conceptual heart of Chapter 3 and Chapter 9.

E. Why no-roll-center force still creates load transfer

Suppose an axle lateral force Y_i acts through its no-roll center Q_i , at height q_i . Because the force acts above the ground, it has a moment about the ground contact line:

$$Y_i q_i.$$

That moment must be balanced by left-right vertical tire-load difference:

$$\Delta Z_i^Y t_i = Y_i q_i.$$

Therefore,

$$\Delta Z_i^Y = \frac{Y_i q_i}{t_i}.$$

This is geometric load transfer.

So:

no suspension roll $\neq$ no lateral load transfer.

This is the most common misunderstanding.

F. Force decomposition through the no-roll axis

Let the total lateral force be Y , acting at height h . We can replace that force by:

Y_1 through Q_1 ,

Y_2 through Q_2 ,

plus a remaining roll moment.

Define the effective no-roll-axis height q^b by

$$Y q^b = Y_1 q_1 + Y_2 q_2.$$

So

$$q^b = \frac{Y_1 q_1 + Y_2 q_2}{Y}.$$

Using

$$Y_1 = \frac{Y a_2 + N_Y}{l}, \quad Y_2 = \frac{Y a_1 - N_Y}{l},$$

we get the errata-consistent formula

$$q^b = \frac{a_2 q_1 + a_1 q_2}{l} + \frac{N_Y}{Y} \frac{q_1 - q_2}{l}$$

or

$$q^b = \frac{a_2 q_1 + a_1 q_2}{l} - \frac{N_Y}{Y} \frac{q_2 - q_1}{l}.$$

The errata explicitly corrects the sign of the N_Y/Y contribution in this part of Chapter 3. SVDerrata

Now the total lateral-force moment about ground is

$$Y h.$$

Split it as

$$Y h = \underbrace{Y q^b}_{\text{geometric/no-roll-center part}} + \underbrace{Y (h - q^b)}_{\text{suspension elastic roll part}}.$$

This is the cleanest way to understand the no-roll axis.

It does not remove load transfer. It decides how much of the load-transfer moment goes through suspension geometry and how much goes through elastic roll stiffness.

G. Suspension and tire roll stiffness

Chapter 3 then says: the remaining roll moment

$$Y (h - q^b)$$

is resisted by suspension and tire compliance.

For axle i :

$k_{\phi i}^s$ = suspension roll stiffness,

$k_{\phi i}^p$ = tire roll stiffness,

$k_{\phi i}$ = combined axle roll stiffness.

The tire roll stiffness is

$$k_{\phi i}^p = \frac{p_i t_i^2}{2}$$

where p_i is the vertical stiffness of one tire.

Suspension and tire roll compliances are in series:

$$k_{\phi i} = \frac{k_{\phi i}^s k_{\phi i}^p}{k_{\phi i}^s + k_{\phi i}^p}.$$

The total vehicle roll stiffness is

$$k_\phi = k_{\phi 1} + k_{\phi 2}.$$

The book explicitly derives these stiffnesses from pure rolling moments and notes that tire vertical stiffness must be separated from suspension roll stiffness. 978-94-017-8533-4

H. Full Chapter 3 lateral-load-transfer formula

The internal equations are:

$$\phi = \phi_1^s + \phi_1^p = \phi_2^s + \phi_2^p,$$

$$Y (h - q^b) = k_{\phi 1}^s \phi_1^s + k_{\phi 2}^s \phi_2^s,$$

$$Y_1 q_1 + k_{\phi 1}^s \phi_1^s = k_{\phi 1}^p \phi_1^p,$$

$$Y_2 q_2 + k_{\phi 2}^s \phi_2^s = k_{\phi 2}^p \phi_2^p.$$

These say:

geometric moment + suspension spring moment = tire vertical-deflection moment.

The errata-corrected suspension-roll deflections are:

$$\phi_1^s = \frac{1}{k_{\phi 1}^s} \frac{k_{\phi 1} k_{\phi 2}}{k_\phi} \left[\frac{Y (h - q^b)}{k_{\phi 2}} - \frac{Y_1 q_1}{k_{\phi 1}^p} + \frac{Y_2 q_2}{k_{\phi 2}^p} \right]$$

$$\phi_2^s = \frac{1}{k_{\phi 2}^s} \frac{k_{\phi 1} k_{\phi 2}}{k_\phi} \left[\frac{Y (h - q^b)}{k_{\phi 1}} - \frac{Y_2 q_2}{k_{\phi 2}^p} + \frac{Y_1 q_1}{k_{\phi 1}^p} \right].$$

The errata is important here because the printed Eq. (3.110) has wrong signs/terms. SVDerrata

Then the lateral load transfers become:

$$\Delta Z_1 = \frac{1}{t_1} \left[\frac{k_{\phi 1}}{k_\phi} Y (h - q^b) + Y_1 q_1 + \frac{k_{\phi 1} k_{\phi 2}}{k_\phi} \left(\frac{Y_2 q_2}{k_{\phi 2}^p} - \frac{Y_1 q_1}{k_{\phi 1}^p} \right) \right]$$

$$\Delta Z_2 = \frac{1}{t_2} \left[\frac{k_{\phi 2}}{k_\phi} Y (h - q^b) + Y_2 q_2 + \frac{k_{\phi 1} k_{\phi 2}}{k_\phi} \left(\frac{Y_1 q_1}{k_{\phi 1}^p} - \frac{Y_2 q_2}{k_{\phi 2}^p} \right) \right].$$

These are the key full Chapter 3 formulas. The book states that the lateral load transfers are linear functions of Y and N_Y , or equivalently of Y_1, Y_2 , and that the simple global equation must still hold. 978-94-017-8533-4

Check:

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = Y (h - q^b) + Y_1 q_1 + Y_2 q_2.$$

But

$$Y_1 q_1 + Y_2 q_2 = Y q^b.$$

Therefore,

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = Y(h - q^b) + Y q^a = Yh.$$

So the detailed suspension result is consistent with global equilibrium.

I. Rigid-tire simplification

If tires are vertically rigid,

$$k_{\phi i}^s \rightarrow \infty,$$

then

$$k_{\phi i} \rightarrow k_{\phi i}^s$$

and the tire-compliance correction terms vanish.

Then

$$k_{\phi} \phi = Y(h - q^b)$$

and

$$\Delta Z_i t_i = Y q_i + k_{\phi i} \phi.$$

This is the version Chapter 9 mostly uses, because Chapter 9 explicitly assumes rigid tires in order to keep the roll-motion theory clear. 978-94-017-8533-4

J. Individual tire loads in Chapter 3

Once $Z_1, Z_2, \Delta Z_1, \Delta Z_2$ are known,

$$Z_{11} = \frac{Z_1}{2} - \Delta Z_1$$

$$Z_{12} = \frac{Z_1}{2} + \Delta Z_1$$

$$Z_{21} = \frac{Z_2}{2} - \Delta Z_2$$

$$Z_{22} = \frac{Z_2}{2} + \Delta Z_2.$$

These tire loads then go back into the tire constitutive equations:

$$F_{xij} = F_{xij}(Z_{ij}, \gamma_{ij}, \sigma_{xij}, \sigma_{yij}),$$

$$F_{yij} = F_{yij}(Z_{ij}, \gamma_{ij}, \sigma_{xij}, \sigma_{yij}).$$

So Chapter 3 is a coupled algebraic-dynamic model:

$$Z_{ij} \rightarrow F_{xij}, F_{yij} \rightarrow X_i, Y_i, \Delta X_i \rightarrow \Delta Z_i \rightarrow Z_{ij}.$$

This coupling is exactly why tire-load equations matter so much in lap simulation and optimal control.

K. How Chapter 3 becomes Chapter 9

Chapter 9 says: Chapter 3 was useful, but it did **not** include true roll/pitch inertial effects. Chapter 3 used roll angle mainly as an internal quasi-static suspension coordinate. Chapter 9 makes roll and pitch actual dynamic coordinates.

So Chapter 9 adds:

$$\begin{array}{ll} \phi = \text{roll angle}, & \theta = \text{pitch angle}, \\ \dot{\phi}, \ddot{\phi}, & \dot{\theta}, \ddot{\theta}. \end{array}$$

It also introduces a more careful point for defining vehicle position: the **Vehicle Invariant Point**, or VIP. The book defines the VIP $\bar{M}$ as the vehicle-body point that, in the reference configuration, coincides with the no-roll center $\bar{Q}$. It uses $\bar{M}$, not G , to define the vehicle lateral velocity when roll exists. 978-94-017-8533-4

This matters because if the car rolls, the center of mass G moves sideways relative to the contact patches. If you used G to define v , your lateral velocity would contain a roll-induced component that is not really “vehicle path” motion.

So Chapter 9 replaces:

$$V_G = u\mathbf{i} + v\mathbf{j}$$

with

$$V_{\bar{M}} = u\mathbf{i}_1 + v\mathbf{j}_1.$$

Then the acceleration of $\bar{M}$ has the same form as Chapter 3:

$$a_{\bar{M}} = (\dot{u} - vr)\mathbf{i}_1 + (\dot{v} + ur)\mathbf{j}_1.$$

The book explicitly says $\bar{M}$ inherits almost everything obtained for G in Chapters 3–7, meaning the Chapter 3 kinematics still largely applies, but now referenced to the VIP instead of the center of mass. 978-94-017-8533-4

L. Chapter 9 equations of motion

For negligible unsprung mass, the translational equations are essentially:

$$m(\dot{u} - vr) + m(h - d)\ddot{\theta} = X$$

$$m(\dot{v} + ur) - m(h - d)\ddot{\phi} = Y$$

$$0 = Z.$$

Here d is the height of the VIP/no-roll reference point, so $h - d$ is the vertical distance from that point to the center of mass.

The rotational equations can be written in the compact alternative form:

$$J_x \ddot{\phi} - J_{xz} \dot{r} - ma_y(h - d) = L_M$$

$$J_y \ddot{\theta} + ma_x(h - d) = M_M$$

$$J_z \dot{r} - J_{zx} \ddot{\phi} + ma_x(h - d)\phi = N_M = N.$$

The book presents these as the roll, pitch, and yaw equations after rewriting the full 3D angular-momentum equations. 978-94-017-8533-4

The errata matters here: it corrects signs in the angular-momentum expressions in Chapter 9. SVDerrata

M. Chapter 9 force and moment right-hand sides

Chapter 9 then supplies the right-hand sides, using almost the same force definitions as Chapter 3:

$$X = X_1 + X_2 - \frac{1}{2} \rho S C_x u^2,$$

$$Y = Y_1 + Y_2,$$

$$Z = Z_1 + Z_1^a + Z_2 + Z_2^a - mg,$$

$$L_M = -k_{\phi} \phi - c_{\phi} \dot{\phi} + mg(h - d)\phi,$$

$$M_M = -k_{\theta} \theta - c_{\theta} \dot{\theta},$$

$$N_M = N = N_Y + N_X = (Y_1 a_1 - Y_2 a_2) + (\Delta X_1 t_1 + \Delta X_2 t_2).$$

The errata corrects the vertical-force equation in Chapter 9: the second aerodynamic vertical-load term must be Z_2^a , not another Z_1^a . SVDerrata

Compare this with Chapter 3:

$$ma_x = X, \quad ma_y = Y, \quad J_z \dot{r} = N.$$

Chapter 9 says those equations were incomplete once roll and pitch have inertia. Now longitudinal acceleration couples to pitch acceleration, lateral acceleration couples to roll acceleration, and yaw couples to roll through J_{zx} .

N. Chapter 9 roll equation versus Chapter 3 roll relation

Start from Chapter 9 roll equation:

$$J_x \ddot{\phi} - J_{xz} \dot{r} - Y(h - d) = L_M.$$

Substitute

$$L_M = -k_{\phi} \phi - c_{\phi} \dot{\phi} + mg(h - d)\phi.$$

Then

$$J_x \ddot{\phi} - J_{xz} \dot{r} - Y(h - d) = -k_{\phi} \phi - c_{\phi} \dot{\phi} + mg(h - d)\phi.$$

Rearrange:

$$Y(h - d) + mg(h - d)\phi = k_{\phi} \phi + c_{\phi} \dot{\phi} + J_x \ddot{\phi} - J_{xz} \dot{r}.$$

This is Chapter 9’s dynamic generalization of the Chapter 3 quasi-static roll relation. The book explicitly says Eq. (9.47) generalizes Eq. (3.117). 978-94-017-8533-4

Compare:

Chapter 3 rigid-tire quasi-static relation:

$$Y(h - d) = k_{\phi} \phi.$$

Chapter 9 dynamic relation:

$$Y(h - d) + mg(h - d)\phi = k_{\phi} \phi + c_{\phi} \dot{\phi} + J_x \ddot{\phi} - J_{xz} \dot{r}.$$

So Chapter 9 adds four things:

$$c_{\phi} \dot{\phi} \quad \text{roll damping,}$$

$$J_x \ddot{\phi} \quad \text{roll inertia,}$$

$$-J_{xz} \dot{r} \quad \text{roll-yaw inertia coupling,}$$

$$mg(h - d)\phi \quad \text{gravity moment due to true body roll.}$$

This is one of the biggest differences between Chapters 3 and 9.

O. Chapter 9 lateral tire-load equations

Chapter 3 rigid-tire formula was:

$$\Delta Z_i t_i = Y_i q_i + k_{\phi i} \phi.$$

Chapter 9 upgrades it to include damping:

$$\Delta Z_1 t_1 = Y_1 q_1 + k_{\phi 1} \phi + c_{\phi 1} \dot{\phi}$$

$$\Delta Z_2 t_2 = Y_2 q_2 + k_{\phi 2} \phi + c_{\phi 2} \dot{\phi}.$$

The book says these are the Chapter 3 load-transfer equations updated because roll is now a time-dependent dynamic variable and shock absorbers must be included. 978-94-017-8533-4

Adding them:

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = k_{\phi} \phi + c_{\phi} \dot{\phi} + Y_1 q_1 + Y_2 q_2.$$

Since

$$Y_1 q_1 + Y_2 q_2 = Y d,$$

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = k_{\phi} \phi + c_{\phi} \dot{\phi} + Y d.$$

Using the dynamic roll equation, Chapter 9 obtains:

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = m a_y h + m g (h - d) \phi - J_x \ddot{\phi} + J_{xz} \dot{r}.$$

This generalizes Chapter 3:

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 \approx m a_y h.$$

So the load transfer now depends explicitly on body angular acceleration. The book explicitly states this as the main difference from Chapter 3. 978-94-017-8533-4

P. Chapter 9 longitudinal load transfer

Chapter 3:

$$\Delta Z \approx - \frac{m a_x h}{l}.$$

Chapter 9:

$$\Delta Z = - \frac{m a_x h + J_y \ddot{\theta}}{l}.$$

So pitch angular acceleration changes the front/rear axle load split.

This is the longitudinal analogue of the roll-inertia correction in lateral load transfer. Chapter 9 explicitly gives this as the generalization of Chapter 3's longitudinal load transfer. 978-94-017-8533-4

Q. Chapter 9 individual tire loads

The final tire-load construction is still Chapter 3's structure:

$$Z_{11} = \frac{Z_1}{2} - \Delta Z_1, \quad Z_{12} = \frac{Z_1}{2} + \Delta Z_1,$$

$$Z_{21} = \frac{Z_2}{2} - \Delta Z_2, \quad Z_{22} = \frac{Z_2}{2} + \Delta Z_2.$$

But now:

$$Z_1 = Z_1^0 + \Delta Z - Z_1^a,$$

$$Z_2 = Z_2^0 - \Delta Z - Z_2^a,$$

with

$$\Delta Z = - \frac{m a_x h + J_y \ddot{\theta}}{l},$$

and

$$\Delta Z_1 t_1 = Y_1 q_1 + k_{\phi 1} \phi + c_{\phi 1} \dot{\phi},$$

$$\Delta Z_2 t_2 = Y_2 q_2 + k_{\phi 2} \phi + c_{\phi 2} \dot{\phi}.$$

So Chapter 9 tire loads are Chapter 3 tire loads with:

pitch inertia added to longitudinal load transfer

and

roll state, roll damping, roll inertia, and yaw-roll coupling added to lateral load transfer.

R. Chapter 3 tire equations still apply in Chapter 9

Chapter 9 explicitly says that taking roll and pitch motion into account does **not directly affect the tire constitutive equations**, so the Chapter 3 tire-force analysis still applies. 978-94-017-8533-4

So the tire model remains:

$$F_{xij} = F_{xij}(Z_{ij}, \gamma_{ij}, \sigma_{xij}, \sigma_{yij}),$$

$$F_{yij} = F_{yij}(Z_{ij}, \gamma_{ij}, \sigma_{xij}, \sigma_{yij}).$$

What changes is how Z_{ij} , γ_{ij} , and possibly the tire velocities are computed.

Chapter 9 says the Chapter 3 congruence equations can mostly be used, but the lateral velocities of the front and rear axes should include small roll-rate corrections:

$$v_1 = v + r a_1 + (d - q_1) \dot{\phi}$$

$$v_2 = v - r a_2 + (d - q_2) \dot{\phi}.$$

The book then says these extra terms are usually very small and can often be neglected. 978-94-017-8533-4

S. The final conceptual bridge

The relationship is:

Chapter 3 gives the algebraic/quasi-static load-transfer closure.

Chapter 9 turns roll and pitch into actual dynamic DOFs.

More explicitly:

Topic	Chapter 3	Chapter 9
Vehicle position	usually G	VIP M
Lateral acceleration	$a_y = \dot{v} + ur$	same form, but for VIP
Roll	internal quasi-static suspension coordinate	real dynamic coordinate $\phi(t)$
Pitch	mostly absent from dynamic model	real dynamic coordinate $\theta(t)$
Tire equations	constitutive equations from slips and loads	same tire equations
Longitudinal load transfer	$\Delta Z \approx -m a_x h / l$	$\Delta Z = (-m a_x h + J_y \ddot{\theta}) / l$
Lateral load transfer	$\Delta Z_i t_i = Y_i q_i + k_{\phi i} \phi$, or full tire-compliance formula	$\Delta Z_i t_i = Y_i q_i + k_{\phi i} \phi + c_{\phi i} \dot{\phi}$
Total lateral transfer	$\Delta Z_1 t_1 + \Delta Z_2 t_2 \approx m a_y h$	includes $m g (h - d) \phi$, $J_x \ddot{\phi}$, $J_{xz} \dot{r}$
No-roll axis	force-decomposition/load-transfer tool	still valid as no-roll axis, but not a physical roll axis

The deepest lesson is this:

The no-roll axis is not the axis about which the car rolls.

It is a geometric force-transmission concept. Chapter 3 uses it to split lateral load transfer into geometric and elastic parts. Chapter 9 then shows that once the body truly rolls, talking about "the axis about which the vehicle rolls" is physically misleading: the VIP/no-roll geometry is useful for defining vehicle motion and load transfer, but not as a literal hinge line. The Chapter 9 summary explicitly says the roll-axis-as-rotation-axis idea is nonsense, while the no-roll-axis concept remains valid. 978-94-017-8533-4

For your own modeling, the clean implementation logic is:

$(u, v, r, \phi, \theta, \dot{\phi}, \dot{\theta}) \rightarrow$ tire slips $\rightarrow F_{xij}, F_{yij} \rightarrow Y_i, \Delta X_i \rightarrow$ EOM $\rightarrow \Delta Z_i, \Delta Z_i \rightarrow Z_{ij} \rightarrow$ tire forces again.

That is the coupled loop Chapter 3 establishes and Chapter 9 upgrades.

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Please double check your above answers, make sure the explanations are full and correct. Make sure your explanations on Chapter 3 and how it relates to Chapter 9 are self-containing and consistent and good enough for me to learn those Chapters.

已思考 2m 12s ›

After double-checking against the book and errata, my earlier explanations were mostly correct, but I would strengthen and slightly refine them in several important places before treating them as your "learning version" for Chapters 3 and 9.

The two biggest refinements are:

- Chapter 3's no-roll-axis derivation must be read with the errata. In particular, the sign in the expanded q_i^0 formula and the printed Eq.(3.110) need correction. The errata explicitly corrects several Chapter 3 equations, including the no-roll/load-transfer-related formulas. SV06-mk
- Chapter 9 uses Chapter 3, but it is not just Chapter 3 with extra symbols. Chapter 3 treats roll mainly as a quasi-static suspension/tire internal coordinate. Chapter 9 makes roll and pitch into actual dynamic coordinates with $\phi, \dot{\phi}, \theta, \dot{\theta}$. The book explicitly says Chapter 9 extends Chapter 3 by taking roll and pitch into account, assumes rigid tires for clarity, and then shows that load transfers depend on angular accelerations. 978-94-017-8533-4 978-94-017-8533-4

Below is the self-contained corrected picture.

1. What Chapter 3 is actually building

Chapter 3 is not only about load transfer. It builds a complete first-order vehicle model for handling and performance. Its structure is:

vehicle kinematics → tire slips → tire forces → global equilibrium → suspension/tire load-transfer closure → individual tire vertical loads.

The contents of Chapter 3 confirm this order: kinematic equations, tire constitutive equations, equilibrium equations, load transfers, suspension first-order analysis, no-roll centers/axis, and finally the complete vehicle model. 978-94-017-8533-4

The key point is that Chapter 3 has two different layers:

global vehicle equilibrium

and

suspension/tire internal equilibrium.

Global equilibrium gives total forces and total moments. But it does **not** by itself tell you how the vertical tire loads are distributed among the four tires.

That missing part is exactly why the no-roll centers and no-roll axis are introduced.

2. Chapter 3 planar kinematics

The vehicle has body-fixed axes:

$\mathbf{i}$ = forward, $\mathbf{j}$ = left/lateral, $\mathbf{k}$ = up.

The velocity of the center of mass G is

$$\mathbf{V}_G = u\mathbf{i} + v\mathbf{j},$$

and the yaw angular velocity is

$$\Omega = r\mathbf{k}.$$

Because $\mathbf{i}, \mathbf{j}$ rotate with yaw rate r ,

$$\dot{\mathbf{i}} = r\mathbf{j}, \quad \dot{\mathbf{j}} = -r\mathbf{i}.$$

Therefore,

$$\begin{aligned} \mathbf{a}_G &= \frac{d}{dt}(u\mathbf{i} + v\mathbf{j}) \\ &= \dot{u}\mathbf{i} + u\dot{\mathbf{i}} + \dot{v}\mathbf{j} + v\dot{\mathbf{j}} \\ &= \dot{u}\mathbf{i} + ur\mathbf{j} + \dot{v}\mathbf{j} - vr\mathbf{i}. \end{aligned}$$

So

$$\mathbf{a}_G = (\dot{u} - vr)\mathbf{i} + (\dot{v} + ur)\mathbf{j}$$

and Chapter 3 defines

$$a_x = \dot{u} - vr$$

$$a_y = \dot{v} + ur.$$

This is the first foundation. Later, the required total longitudinal and lateral road forces are

$$X = ma_x, \quad Y = ma_y.$$

3. Tire forces in the vehicle frame

Each tire produces forces in its own wheel frame:

$$F_{xij}, \quad F_{yij}.$$

But the vehicle equilibrium equations need forces in the vehicle frame:

$$X_{ij} = F_{xij} \cos \delta_{ij} - F_{yij} \sin \delta_{ij},$$

$$Y_{ij} = F_{xij} \sin \delta_{ij} + F_{yij} \cos \delta_{ij}.$$

Then define axle sums:

$$X_1 = X_{11} + X_{12}, \quad X_2 = X_{21} + X_{22},$$

$$Y_1 = Y_{11} + Y_{12}, \quad Y_2 = Y_{21} + Y_{22}.$$

Also define left-right half-differences:

$$\Delta X_1 = \frac{X_{12} - X_{11}}{2}, \quad \Delta X_2 = \frac{X_{22} - X_{21}}{2},$$

$$\Delta Z_1 = \frac{Z_{12} - Z_{11}}{2}, \quad \Delta Z_2 = \frac{Z_{22} - Z_{21}}{2}.$$

This notation is essential because ΔX_i contributes to yaw moment, while ΔZ_i is the lateral load transfer at axle i .

4. Chapter 3 global in-plane equilibrium

The main in-plane equations are:

$$m(\dot{u} - vr) = X_1 + X_2 - X_a$$

$$m(\dot{v} + ur) = Y_1 + Y_2$$

$$J_z \dot{r} = Y_1 a_1 - Y_2 a_2 + \Delta X_1 t_1 + \Delta X_2 t_2.$$

Define

$$Y = Y_1 + Y_2,$$

$$N_X = \Delta X_1 t_1 + \Delta X_2 t_2,$$

$$N_Y = Y_1 a_1 - Y_2 a_2.$$

Then

$$J_z \dot{r} = N_Y + N_X,$$

so

$$N_Y = J_z \dot{r} - N_X.$$

Now solve

$$Y_1 + Y_2 = Y,$$

$$Y_1 a_1 - Y_2 a_2 = N_Y.$$

From the first equation,

$$Y_2 = Y - Y_1.$$

Substitute into the yaw-moment equation:

$$Y_1 a_1 - (Y - Y_1) a_2 = N_Y,$$

$$Y_1 a_1 - Y a_2 + Y_1 a_2 = N_Y,$$

$$Y_1 (a_1 + a_2) = Y a_2 + N_Y.$$

Since

$$l = a_1 + a_2,$$

we get

$$Y_1 = \frac{Y a_2 + N_Y}{l}$$

and

$$Y_2 = \frac{Y a_1 - N_Y}{l}.$$

This is extremely important:

global force/yaw equilibrium determines Y_1, Y_2 .

Suspension geometry does not choose Y_1, Y_2 . Suspension geometry chooses how the corresponding vertical load transfer is distributed.

5. Chapter 3 longitudinal load transfer

The vertical axle loads are

$$Z_1 = Z_{11} + Z_{12}, \quad Z_2 = Z_{21} + Z_{22}.$$

The vertical force balance is, with the errata correction on aerodynamic loads,

$$Z_1 + Z_1^a + Z_2 + Z_2^a = mg.$$

The pitch-moment balance gives

$$(Z_1 + Z_1^a) a_1 - (Z_2 + Z_2^a) a_2 = -ma_x h + J_{zz} r^2.$$

Solving gives

$$Z_1 = Z_1^0 + \Delta Z - Z_1^a$$

$$Z_2 = Z_2^0 - \Delta Z - Z_2^a$$

where

$$Z_1^0 = \frac{mga_2}{l}$$

$$Z_2^0 = \frac{mga_1}{l}$$

and

$$\Delta Z = -\frac{ma_x h}{l} + \frac{J_{zz} r^2}{l} \approx -\frac{ma_x h}{l}.$$

So:

- braking: $a_x < 0 \Rightarrow \Delta Z > 0$, front axle gains load;
- acceleration: $a_x > 0 \Rightarrow \Delta Z < 0$, rear axle gains load.

This part does not need the no-roll axis. Longitudinal load transfer comes from longitudinal acceleration acting through the height h .

6. Chapter 3 global lateral load transfer

The global roll-moment balance gives

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = Yh + J_{xx} \ddot{\phi}.$$

Usually $J_{xx} \ddot{\phi}$ is small, so

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 \approx Yh = m a_y h.$$

This equation is central, but it is incomplete. It is one scalar equation for two unknowns:

$$\Delta Z_1, \quad \Delta Z_2.$$

Therefore:

global equilibrium gives total lateral load transfer, not the front/rear split.

This is why Chapter 3 needs suspension analysis.

7. What the no-roll center really means

For each axle i , Chapter 3 constructs a point Q_i , height q_i , from suspension geometry.

For a symmetric independent suspension, geometrically:

- take the tire contact point A_i ;
- find the instantaneous center B_i of the wheel hub relative to the body;
- draw the lines $A_i B_i$ for left and right sides;
- their intersection on the vehicle center plane is Q_i .

The front and rear points are

$$Q_1, \quad Q_2.$$

Their heights are

$$q_1, \quad q_2.$$

The line through Q_1, Q_2 is traditionally called the roll axis. Guiggiani argues that this name is misleading and calls it the **no-roll axis**.

The correct meaning is:

a lateral force through Q_i produces no suspension-spring roll at axle i .

It does **not** mean:

the vehicle body physically rolls about Q_i .

It does **not** mean:

there is no load transfer.

It only means:

the suspension springs are not the path creating the roll moment.

A force through Q_i can still create:

geometric load transfer,

tire vertical deflection,

suspension jacking,

body vertical displacement.

This is the conceptual heart of Chapter 3.

8. Why a force through the no-roll center still creates load transfer

Suppose a lateral axle force Y_i acts through Q_i , height q_i .

Even though this force does not produce suspension-spring roll, it is still applied at height q_i above the ground. Therefore, it produces a moment

$$Y_i q_i.$$

That moment must be balanced by a left-right vertical tire-load difference:

$$\Delta Z_i^Y t_i = Y_i q_i.$$

So

$$\Delta Z_i^Y = \frac{Y_i q_i}{t_i}.$$

This is the **geometric load-transfer** part.

Therefore:

no suspension roll $\neq$ no load transfer.

This sentence is probably the most important idea in Chapter 3.

9. Force decomposition through the no-roll axis

Let the total lateral force be

$$Y = Y_1 + Y_2.$$

In the global vehicle model, that force acts at height h . Chapter 3 decomposes it into:

Y_1 through Q_1 ,

Y_2 through Q_2 ,

plus a pure roll moment.

Define the effective no-roll-axis height q^b by

$$Y q^b = Y_1 q_1 + Y_2 q_2.$$

So

$$q^b = \frac{Y_1 q_1 + Y_2 q_2}{Y}.$$

Using

$$Y_1 = \frac{Y a_2 + N_Y}{l}, \quad Y_2 = \frac{Y a_1 - N_Y}{l},$$

we get

$$q^b = \frac{1}{Y} \left[\frac{Y a_2 + N_Y}{l} q_1 + \frac{Y a_1 - N_Y}{l} q_2 \right].$$

So

$$q^b = \frac{a_2 q_1 + a_1 q_2}{l} + \frac{N_Y}{Y} \frac{q_1 - q_2}{l}.$$

Equivalently,

$$q^b = \frac{a_2 q_1 + a_1 q_2}{l} - \frac{N_Y}{Y} \frac{q_2 - q_1}{l}.$$

This is the errata-consistent **sign**. The printed Chapter 3 expression must be corrected according to the errata. sv06rara

Now split the total lateral-force moment:

$$Yh = Y q^b + Y(h - q^b).$$

So:

$$Y q^b = Y_1 q_1 + Y_2 q_2$$

is the geometric/no-roll-center part, and

$$Y(h - q^b)$$

is the remaining roll moment that must be resisted by suspension/tire roll compliance.

This is the cleanest interpretation of the no-roll axis:

the no-roll axis splits Yh into a geometric part and an elastic-roll part.

It does not remove load transfer.

10. Suspension and tire roll stiffness in Chapter 3

For axle i , Chapter 3 separates:

$k_{\phi i}^s$ = suspension roll stiffness,

$k_{\phi i}^p$ = tire roll stiffness,

$k_{\phi i}$ = combined axle roll stiffness.

Because suspension and tire roll compliances are in series,

$$k_{\phi i} = \frac{k_{\phi i}^s k_{\phi i}^p}{k_{\phi i}^s + k_{\phi i}^p}.$$

The tire roll stiffness is

$$k_{\phi i}^p = \frac{p_i t_i^2}{2},$$

where p_i is the vertical stiffness of one tire on axle i .

The total vehicle roll stiffness is

$$k_{\phi} = k_{\phi 1} + k_{\phi 2}.$$

This is another important point: in Chapter 3, tire vertical compliance is not a minor detail. The book explicitly warns that for formula cars it may be unsafe to treat tires as vertically rigid. 978-94-017-8533-4

11. Chapter 3 internal equilibrium equations

At each axle,

$$\phi = \phi_i^s + \phi_i^p,$$

where:

ϕ_i^s = roll due to suspension deflection,

$$\phi_l^P = \text{roll due to tire vertical deflection.}$$

The remaining elastic roll moment is

$$Y(h - q^b).$$

So

$$Y(h - q^b) = k_{\phi 1}^s \phi_1^s + k_{\phi 2}^s \phi_2^s.$$

At each axle, the total load-transfer moment is

$$\Delta Z_i t_i = Y_i q_i + k_{\phi i}^s \phi_i^s = k_{\phi i}^P \phi_i^P.$$

This equation is the core of the Chapter 3 load-transfer derivation.

It says:

$$\text{total lateral load transfer} = \text{geometric no-roll-center part} + \text{suspension elastic part.}$$

The same total load-transfer moment also equals the tire vertical-deflection moment.

12. Corrected Chapter 3 lateral-load-transfer formulas

Solving the internal system gives the full lateral load-transfer formulas:

$$\begin{aligned} \Delta Z_1 &= \frac{1}{t_1} \left[k_{\phi}^{s1} Y(h - q^b) + Y_1 q_1 + \frac{k_{\phi 1} k_{\phi 2}}{k_{\phi}} \left(\frac{Y_2 q_2}{k_{\phi 2}^P} - \frac{Y_1 q_1}{k_{\phi 1}^P} \right) \right] \\ \Delta Z_2 &= \frac{1}{t_2} \left[k_{\phi}^{s2} Y(h - q^b) + Y_2 q_2 + \frac{k_{\phi 1} k_{\phi 2}}{k_{\phi}} \left(\frac{Y_1 q_1}{k_{\phi 1}^P} - \frac{Y_2 q_2}{k_{\phi 2}^P} \right) \right]. \end{aligned}$$

Interpretation:

$$\frac{k_{\phi i}}{k_{\phi}} Y(h - q^b)$$

is the elastic roll-stiffness distribution part.

$$Y_i q_i$$

is the geometric no-roll-center part.

The terms involving $k_{\phi i}^P$ are the correction caused by tire vertical compliance.

Now check consistency:

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = Y(h - q^b) + Y_1 q_1 + Y_2 q_2.$$

But

$$Y_1 q_1 + Y_2 q_2 = Y q^b.$$

Therefore,

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = Y(h - q^b) + Y q^b = Yh.$$

So the detailed suspension formulas are consistent with global roll-moment equilibrium.

That is the proof you should always keep in mind.

13. Rigid-tire simplification in Chapter 3

If tire vertical deflection is neglected,

$$k_{\phi i}^P \rightarrow \infty,$$

so

$$k_{\phi i} \rightarrow k_{\phi i}^s.$$

Then

$$k_{\phi} \phi = Y(h - q^b)$$

and

$$\Delta Z_i t_i = Y_i q_i + k_{\phi i} \phi.$$

Equivalently,

$$\Delta Z_i t_i = \frac{k_{\phi i}}{k_{\phi}} Y(h - q^b) + Y_i q_i.$$

This is much easier to remember, but it is less complete than the tire-compliance version.

This is also the version most directly connected to Chapter 9, because Chapter 9 explicitly assumes rigid tires to avoid making the theory too involved.
978-94-017-8533-4

14. Final Chapter 3 tire vertical loads

Once Z_1 , Z_2 , ΔZ_1 , ΔZ_2 are known,

$$Z_{11} = \frac{Z_1}{2} - \Delta Z_1$$

$$Z_{12} = \frac{Z_1}{2} + \Delta Z_1$$

$$Z_{21} = \frac{Z_2}{2} - \Delta Z_2$$

$$Z_{22} = \frac{Z_2}{2} + \Delta Z_2.$$

Substituting the longitudinal axle loads:

$$Z_1 = Z_1^0 + \Delta Z - Z_1^a,$$

$$Z_2 = Z_2^0 - \Delta Z - Z_2^a.$$

Therefore,

$$Z_{11} = \frac{1}{2} \left(Z_1^0 + \Delta Z - Z_1^a \right) - \Delta Z_1$$

$$Z_{12} = \frac{1}{2} \left(Z_1^0 + \Delta Z - Z_1^a \right) + \Delta Z_1$$

$$Z_{21} = \frac{1}{2} \left(Z_2^0 - \Delta Z - Z_2^a \right) - \Delta Z_2$$

$$Z_{22} = \frac{1}{2} \left(Z_2^0 - \Delta Z - Z_2^a \right) + \Delta Z_2.$$

These vertical loads then feed the tire equations:

$$F_{xij} = F_{xij}(Z_{ij}, \gamma_{ij}, \sigma_{xij}, \sigma_{yij}, \varphi_{ij}),$$

$$F_{yij} = F_{yij}(Z_{ij}, \gamma_{ij}, \sigma_{xij}, \sigma_{yij}, \varphi_{ij}).$$

So Chapter 3 is a coupled loop:

$$Z_{ij} \rightarrow F_{xij}, F_{yij} \rightarrow X_i, Y_i, \Delta X_i \rightarrow \Delta Z_i \rightarrow Z_{ij}.$$

That loop is exactly why tire vertical load computation is not optional in a serious handling model.

15. What my previous explanation did not emphasize enough

The previous explanations were correct in the main direction, but I should have emphasized these learning cautions more strongly.

First, the no-roll axis is a **force-decomposition concept**, not a physical rotation axis. Chapter 9 strongly reinforces this and says the roll-axis-as-rotation-axis idea is meaningless, while the no-roll-axis concept remains valid.
978-94-017-8533-4

Second, q^b is not merely “some average roll-center height.” It is the height of the point on the no-roll axis relevant to the current lateral force decomposition:

$$q^b = \frac{Y_1 q_1 + Y_2 q_2}{Y}.$$

If Y_1/Y_2 changes, q^b can change. If $q_1 \approx q_2$, then q^b is almost constant.

Third, Chapter 3’s full formula includes tire vertical compliance. Chapter 9 deliberately simplifies by assuming rigid tires. Therefore, if you want to build a more faithful model than Chapter 9’s simplified theory, you should not blindly discard the Chapter 3 tire-compliance terms.

Fourth, the errata matters. For learning, use the corrected q^b , corrected Eq. (3.110), corrected rear tire-slip signs, corrected tire friction inequality, and corrected Chapter 9 vertical aerodynamic load expression.

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16. How Chapter 3 relates to Chapter 9

Now the bridge.

Chapter 3 says:

we need roll angle to compute load transfer, but we will not treat roll as a real dynamic DOF.

Chapter 9 says:

now roll and pitch are actual dynamic motions of the vehicle body.

So Chapter 9 keeps the Chapter 3 architecture, but modifies the equations of motion and the load-transfer equations.

17. Chapter 9 changes the vehicle reference point

In Chapter 3, the vehicle lateral velocity was defined at G :

$$V_G = u\mathbf{i} + v\mathbf{j}.$$

In Chapter 9, this becomes problematic, because when the car rolls, G moves laterally relative to the tire contact patches. That lateral motion is caused by roll, not by the vehicle path.

Therefore Chapter 9 introduces the **Vehicle Invariant Point**, or VIP, denoted $\bar{M}$.

Chapter 9 defines the vehicle velocity as the velocity of $\bar{M}$:

$$\mathbf{V}_M = u\hat{\mathbf{i}}_1 + v\hat{\mathbf{j}}_1.$$

Then

$$\mathbf{a}_M = (\dot{u} - vr)\hat{\mathbf{i}}_1 + (\dot{v} + ur)\hat{\mathbf{j}}_1.$$

This is intentionally almost identical to Chapter 3’s planar acceleration formula, but now it is attached to the VIP rather than G . The book says M inherits almost everything obtained for G in Chapters 3–7.

978-94-017-8533-4

So:

Chapter 3 kinematics survive in Chapter 9, but the reference point changes from G to M .

18. Chapter 9 orientation: yaw, pitch, roll

Chapter 9 introduces yaw-pitch-roll rotations:

$$\psi = \text{yaw}, \qquad \theta = \text{pitch}, \qquad \phi = \text{roll}.$$

For small pitch and roll, the body angular velocity components are approximately

$$p \simeq \dot{\phi} - r\theta,$$

$$q \simeq \dot{\theta} + r\phi,$$

$$r \simeq \dot{\psi}.$$

This matters because the roll and pitch equations are not just simple scalar spring equations. They come from 3D rigid-body angular momentum.

The errata also corrects Chapter 9 angular-momentum expressions, so the corrected version should be used if you derive these equations carefully from the book. SVDerrata

19. Chapter 9 equations of motion: simplified no-unsprung-mass version

For the simplest Chapter 9 case, neglecting unsprung mass, the translational equations are:

$$m \left[(\dot{u} - vr) + (h - d)\ddot{\theta} \right] = X$$

$$m \left[(\dot{v} + ur) - (h - d)\dot{\phi} \right] = Y$$

$$0 = Z.$$

Here d is the height associated with the no-roll/VIP reference point. Conceptually, $h - d$ is the vertical distance from the VIP/no-roll reference point to the center of mass.

Compare with Chapter 3:

$$m(\dot{u} - vr) = X,$$

$$m(\dot{v} + ur) = Y.$$

Chapter 9 adds:

$$m(h - d)\ddot{\theta}$$

to the longitudinal equation and

$$-m(h - d)\dot{\phi}$$

to the lateral equation.

That means:

pitch acceleration affects longitudinal force balance;

roll acceleration affects lateral force balance.

The rotational equations, in the compact form, are:

$$J_x\ddot{\phi} - J_{xz}\dot{r} - ma_y(h - d) = L_M$$

$$J_y\ddot{\theta} + ma_x(h - d) = M_M$$

$$J_z\dot{r} - J_{xz}\dot{\phi} + ma_x(h - d)\phi = N_M = N.$$

These are the dynamic generalization of Chapter 3’s out-of-plane equilibrium and yaw equation.

20. Chapter 9 force and moment right-hand sides

Chapter 9 uses nearly the same tire-force definitions as Chapter 3:

$$X = X_1 + X_2 - \frac{1}{2}\rho SC_x u^2,$$

$$Y = Y_1 + Y_2,$$

$$N = N_Y + N_X = (Y_1a_1 - Y_2a_2) + (\Delta X_1t_1 + \Delta X_2t_2).$$

The vertical force equation must be read with the errata correction:

$$Z = Z_1 + Z_1^* + Z_2 + Z_2^* - mg.$$

The printed Chapter 9 equation repeats Z_1^* ; the errata corrects the second one to Z_2^* . SVDerrata

The roll and pitch moments are:

$$L_M = -k_\phi\phi - c_\phi\dot{\phi} + mg(h - d)\phi$$

$$M_M = -k_\theta\theta - c_\theta\dot{\theta}.$$

So Chapter 9 adds roll damping and pitch damping:

$$c_\phi\dot{\phi}, \qquad c_\theta\dot{\theta}.$$

21. Chapter 9 roll equation versus Chapter 3 roll equation

From Chapter 9:

$$J_x\ddot{\phi} - J_{xz}\dot{r} - ma_y(h - d) = L_M.$$

Substitute

$$L_M = -k_\phi\phi - c_\phi\dot{\phi} + mg(h - d)\phi.$$

Then

$$J_x\ddot{\phi} - J_{xz}\dot{r} - ma_y(h - d) = -k_\phi\phi - c_\phi\dot{\phi} + mg(h - d)\phi.$$

Move terms:

$$ma_y(h - d) + mg(h - d)\phi = k_\phi\phi + c_\phi\dot{\phi} + J_x\ddot{\phi} - J_{xz}\dot{r}.$$

Since

$$Y = ma_{y1},$$

$$Y(h - d) + mg(h - d)\phi = k_\phi\phi + c_\phi\dot{\phi} + J_x\ddot{\phi} - J_{xz}\dot{r}.$$

This is Chapter 9’s dynamic generalization of Chapter 3’s rigid-tire quasi-static equation:

$$Y(h - q^*) = k_\phi\phi.$$

So Chapter 9 adds:

$$c_\phi\dot{\phi}$$

roll damping,

$$J_x\ddot{\phi}$$

roll inertia,

$$-J_{xz}\dot{r}$$

yaw-roll inertial coupling,

and

$$mg(h - d)\phi$$

the gravity moment due to actual body roll.

This is one of the most important bridges from Chapter 3 to Chapter 9.

22. Chapter 9 lateral tire-load equations

Chapter 3 rigid-tire formula:

$$\Delta Z_1t_i = Y_1q_{1i} + k_{\phi\delta}\phi.$$

Chapter 9 adds roll damping:

$$\Delta Z_1t_1 = Y_1q_1 + k_{\phi 1}\phi + c_{\phi 1}\dot{\phi}$$

$$\Delta Z_2t_2 = Y_2q_2 + k_{\phi 2}\phi + c_{\phi 2}\dot{\phi}.$$

Add them:

$$\Delta Z_1t_1 + \Delta Z_2t_2 = k_\phi\phi + c_\phi\dot{\phi} + Y_1q_1 + Y_2q_2.$$

Using the Chapter 3 identity,

$$Y_1q_1 + Y_2q_2 = Yd,$$

we get

$$\Delta Z_1t_1 + \Delta Z_2t_2 = k_\phi\phi + c_\phi\dot{\phi} + Yd.$$

Then use the dynamic roll equation to eliminate $k_\phi\phi + c_\phi\dot{\phi}$. From above:

$$k_\phi\phi + c_\phi\dot{\phi} = Y(h - d) + mg(h - d)\phi - J_x\ddot{\phi} + J_{xz}\dot{r}.$$

Therefore:

$$\Delta Z_1t_1 + \Delta Z_2t_2 = Y(h - d) + Yd + mg(h - d)\phi - J_x\ddot{\phi} + J_{xz}\dot{r}.$$

Since

$$Y(h-d)+Yd=Yh,$$

$$\Delta Z_1t_1+\Delta Z_2t_2=Yh+mg(h-d)\phi-J_x\ddot{\phi}+J_{xz}\ddot{r}.$$

Because

$$Y=ma_y,$$

$$\Delta Z_1t_1+\Delta Z_2t_2=ma_yh+mg(h-d)\phi-J_x\ddot{\phi}+J_{xz}\ddot{r}.$$

Compare with Chapter 3:

$$\Delta Z_1t_1+\Delta Z_2t_2\approx ma_yh.$$

So Chapter 9's main new lateral-load-transfer insight is:

lateral load transfer depends on roll angle and roll angular acceleration.

That is exactly why Chapter 9 is needed for transient roll behavior.

23. Chapter 9 longitudinal load transfer

Chapter 3:

$$\Delta Z\approx-\frac{ma_xh}{l}.$$

Chapter 9:

$$\Delta Z=\frac{-ma_xh+J_y\ddot{\theta}}{l}.$$

So pitch angular acceleration modifies longitudinal load transfer.

Interpretation:

$$-ma_xh$$

is the usual acceleration/braking load-transfer moment.

$$J_y\ddot{\theta}$$

is the pitch-inertia correction.

Therefore:

Chapter 9 adds pitch dynamics to longitudinal load transfer.

24. Chapter 9 individual tire loads

The final construction is still the Chapter 3 construction:

$$Z_{11}=\frac{Z_1}{2}-\Delta Z_1,\quad Z_{12}=\frac{Z_1}{2}+\Delta Z_1,$$

$$Z_{21}=\frac{Z_2}{2}-\Delta Z_2,\quad Z_{22}=\frac{Z_2}{2}+\Delta Z_2.$$

But now:

$$Z_1=Z_1^0+\Delta Z-Z_1^a,$$

$$Z_2=Z_2^0-\Delta Z-Z_2^a,$$

with

$$\Delta Z=\frac{-ma_xh+J_y\ddot{\theta}}{l}.$$

And the lateral load transfers are:

$$\Delta Z_1=\frac{Y_1q_1+k_{s1}\phi+c_{s1}\dot{\phi}}{t_1}$$

$$\Delta Z_2=\frac{Y_2q_2+k_{s2}\phi+c_{s2}\dot{\phi}}{t_2}.$$

So Chapter 9 tire loads are Chapter 3 tire loads with:

pitch acceleration added to longitudinal load transfer

and

roll state/damping/dynamics added to lateral load transfer.

25. Important nuance about q^h in Chapter 3 and $\bar{d}$ in Chapter 9

This is a subtle but important learning point.

In Chapter 3,

$$q^h=\frac{Y_1q_1+Y_2q_2}{Y}.$$

So q^h is the no-roll-axis height corresponding to the current front/rear lateral-force split.

In Chapter 9, Guiggiani uses $\bar{d}$ because q is already used as an angular-velocity component. For load-transfer algebra, $\bar{d}$ plays the same role as q^h :

$$Y\bar{d}=Y_1q_1+Y_2q_2.$$

So for load-transfer derivations, you can mentally identify

$$\bar{d}\leftrightarrow q^h.$$

But be careful: if Y_1/Y_2 changes strongly because N_Y changes, then q^h changes unless $q_1\approx q_2$. For careful numerical work, you should keep the Chapter 3 expression

$$q^h=\frac{a_2q_1+a_1q_2}{l}=\frac{N_Y}{Y}\frac{q_2-q_1}{l}$$

rather than blindly treating the no-roll height as a universal constant.

26. Chapter 9 tire equations and congruence equations

Chapter 9 says roll and pitch do not directly change the tire constitutive equations. The same tire law structure from Chapter 3 remains:

$$F_{xij}=F_{xij}(Z_{ij},\gamma_{ij},\sigma_{xij},\sigma_{yij},\varphi_{ij}),$$

$$F_{yij}=F_{yij}(Z_{ij},\gamma_{ij},\sigma_{xij},\sigma_{yij},\varphi_{ij}).$$

But roll and pitch can change the inputs to those tire laws:

$$Z_{ij},\quad \gamma_{ij},\quad \sigma_{xij},\quad \sigma_{yij}.$$

Chapter 9 says the Chapter 3 congruence equations can mostly still be used, but the front and rear lateral velocities should, more precisely, include small roll-rate corrections:

$$v_1=v+ra_1+(d-q_1)\dot{\phi}$$

$$v_2=v-ra_2+(d-q_2)\dot{\phi}.$$

The book says these terms are usually very small and can often be neglected. 978-94-017-8533-4

For your modeling work, this means:

Chapter 9 mostly preserves Chapter 3 tire slip structure, but improves the kinematics if roll rate is important.

27. The complete learning map

Here is the cleanest way to learn the two chapters.

Chapter 3:

$$u,v,r\rightarrow a_x,a_y\rightarrow X,Y,N\rightarrow Y_1,Y_2\rightarrow\Delta Z,\Delta Z_1,\Delta Z_2\rightarrow Z_{ij}\rightarrow F_{xij},F_{yij}.$$

Chapter 9:

$$u,v,r,\phi,\dot{\phi},\ddot{\phi},\theta,\dot{\theta},\ddot{\theta}\rightarrow a_x,a_y\rightarrow X,Y,N,L_M,M_M\rightarrow Y_1,Y_2\rightarrow\Delta Z,\Delta Z_1,\Delta Z_2\rightarrow Z_{ij}\rightarrow F_{xij},F_{yij}.$$

So Chapter 9 is Chapter 3 plus:

real roll dynamics

real pitch dynamics

VIP instead of simply using G

angular-acceleration corrections to load transfer.

28. Final corrected conceptual summary

Chapter 3 teaches:

total lateral load transfer comes from Yh .

But:

front/rear lateral load-transfer split comes from suspension geometry and stiffness.

The no-roll axis is used because:

$$Yh=Yq^h+Y(h-q^h).$$

The first part,

$$Yq^h=Y_1q_1+Y_2q_2,$$

is geometric load transfer through the suspension linkages.

The second part,

$$Y(h-q^h),$$

is the elastic roll moment handled by suspension/tire roll stiffness.

Chapter 9 teaches:

the car does not physically roll about the no-roll axis.

Instead, roll is a true 3D rigid-body motion with $\phi, \dot{\phi}, \ddot{\phi}$. Therefore:

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = m a_y h + m g (h - d) \phi - J_x \ddot{\phi} + J_{xz} \dot{r},$$

and

$$\Delta Z = \frac{-m a_x h + J_y \ddot{\theta}}{l}.$$

So the deepest relationship is:

Chapter 3 gives the quasi-static load-transfer closure.

Chapter 9 turns that closure into a dynamic roll/pitch model.

For understanding Chapter 9, the most important Chapter 3 ideas are not the long tire-slip formulas. They are these:

Y_1, Y_2 come from global force/yaw equilibrium.

$\Delta Z_1, \Delta Z_2$ require suspension/tire internal equilibrium.

no-roll center means no suspension-spring roll, not no load transfer.

no-roll axis is not a physical roll axis.

individual tire loads close the loop between vehicle dynamics and tire forces.

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Check throughout this conversation between you and me, make sure you have provided all the necessary information for me to learn Chapter 3 of the book.

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I checked the whole thread. For your original focus — Chapter 3’s lateral load computation, no-roll center/no-roll axis, and why Chapter 3 is the foundation of Chapter 9 — we have covered the necessary core ideas correctly.

But for “fully learn Chapter 3” as a whole, the answer is: not completely yet. We have explained the central load-transfer/no-roll-axis material very thoroughly, but Chapter 3 also contains other pieces that you should understand so the chapter becomes a complete model, not only a load-transfer derivation. The chapter’s own structure includes kinematics, tire slips, tire constitutive equations, global equilibrium, load transfers, suspension first-order analysis, dependent suspensions, sprung/unsprung masses, and the final assembled handling/performance model. 978-94-017-8533-4

So the best correction is:

We have covered the most difficult and most important part, but not every necessary part.

Below is the checklist of what is already covered and what still needs to be added.

1. What has already been covered well

1.1 Planar acceleration kinematics

We derived:

$$\begin{aligned} V_G &= u\mathbf{i} + v\mathbf{j} \\ \Omega &= r\mathbf{k} \\ a_G &= (\dot{u} - vr)\mathbf{i} + (\dot{v} + ur)\mathbf{j} \end{aligned}$$

so

$$a_x = \dot{u} - vr, \quad a_y = \dot{v} + ur.$$

This is correct and essential.

1.2 Global force and yaw equilibrium

We covered:

$$\begin{aligned} m(\dot{u} - vr) &= X_1 + X_2 - X_a \\ m(\dot{v} + ur) &= Y_1 + Y_2 \\ J_z \dot{r} &= Y_1 a_1 - Y_2 a_2 + \Delta X_1 t_1 + \Delta X_2 t_2. \end{aligned}$$

Then:

$$\begin{aligned} Y &= Y_1 + Y_2 \\ N_X &= \Delta X_1 t_1 + \Delta X_2 t_2 \\ N_Y &= Y_1 a_1 - Y_2 a_2 \\ N_Y &= J_z \dot{r} - N_X. \end{aligned}$$

Solving gives:

$$\begin{aligned} Y_1 &= \frac{Y a_2 + N_Y}{l} \\ Y_2 &= \frac{Y a_1 - N_Y}{l}. \end{aligned}$$

This is correct. It is also one of the most important Chapter 3 conclusions: Y_1, Y_2 come from global equilibrium, not from suspension geometry.

The book explicitly emphasizes that the front/rear lateral forces are obtained from global equilibrium and are not affected by suspension type, tire type, or grip level in that algebraic sense. 978-94-017-8533-4

1.3 Longitudinal load transfer

We covered:

$$\begin{aligned} Z_1 &= Z_1^0 + \Delta Z - Z_1^a \\ Z_2 &= Z_2^0 - \Delta Z - Z_2^a \end{aligned}$$

with

$$Z_1^0 = \frac{m g a_2}{l}, \quad Z_2^0 = \frac{m g a_1}{l}$$

and

$$\Delta Z = -\frac{m a_x h}{l} + \frac{J_{xz} r^2}{l} \approx -\frac{m a_x h}{l}.$$

This is correct. The book gives this directly in the longitudinal load-transfer section. 978-94-017-8533-4

1.4 Global lateral load-transfer equation

We covered:

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = Y h + J_{xz} \dot{r}$$

and usually

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 \approx Y h = m a_y h.$$

This is correct.

The important point is:

this gives total lateral load transfer, not front/rear distribution.

The book explicitly says one equation is not enough to obtain $\Delta Z_1, \Delta Z_2$, and that suspension geometry and compliance directly influence $\Delta Z_1 / \Delta Z_2$. 978-94-017-8533-4

1.5 No-roll center and no-roll axis

We covered the correct conceptual interpretation:

no-roll center $\neq$ physical roll hinge.

A no-roll center is the point through which a lateral force can be transmitted without producing **suspension-spring roll** at that axle.

But that same force can still produce:

geometric load transfer,
tire vertical deflection,
suspension jacking,
body vertical displacement.

This is correct and essential.

We also correctly emphasized:

no suspension roll $\neq$ no load transfer.

1.6 Force decomposition through no-roll axis

We covered:

$$Y q^b = Y_1 q_1 + Y_2 q_2$$

so

$$q^b = \frac{Y_1 q_1 + Y_2 q_2}{Y}.$$

Using the front/rear lateral force split,

$$q^b = \frac{a_2 q_1 + a_1 q_2}{l} + \frac{N_Y}{Y} \frac{q_1 - q_2}{l}$$

or

$$q^b = \frac{a_2 q_1 + a_1 q_2}{l} - \frac{N_Y}{Y} \frac{q_2 - q_1}{l}.$$

This is the errata-consistent sign. The errata corrects several Chapter 3 equations, including no-roll/load-transfer-related equations and the later tire-slip formulas. SV0Errata

Then:

$$Yh = Yq^h + Y(h - q^h).$$

This is the cleanest no-roll-axis interpretation:

$$Yq^h = \text{geometric load-transfer moment}$$

$$Y(h - q^h) = \text{suspension/tire elastic roll moment.}$$

1.7 Full lateral load transfer with tire vertical compliance

We covered the full Chapter 3 structure:

$$\Delta Z_i t_i = Y_{q_i} + k_{\phi i}^s \phi_i^s = k_{\phi i}^p \phi_i^p.$$

And the full formulas:

$$\begin{aligned}\Delta Z_1 &= \frac{1}{t_1} \left[k_{\phi}^{o1} Y(h - q^h) + Y_1 q_1 + \frac{k_{\phi 1} k_{\phi 2}}{k_{\phi}} \left(\frac{Y_2 q_2}{k_{\phi 2}^p} - \frac{Y_1 q_1}{k_{\phi 1}^p} \right) \right] \\ \Delta Z_2 &= \frac{1}{t_2} \left[k_{\phi}^{o2} Y(h - q^h) + Y_2 q_2 + \frac{k_{\phi 1} k_{\phi 2}}{k_{\phi}} \left(\frac{Y_1 q_1}{k_{\phi 1}^p} - \frac{Y_2 q_2}{k_{\phi 2}^p} \right) \right].\end{aligned}$$

This is correct.

We also checked consistency:

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = Y(h - q^h) + Y_1 q_1 + Y_2 q_2 = Y(h - q^h) + Yq^h = Yh.$$

This consistency check is very important.

1.8 Rigid-tire simplification

We covered:

$$k_{\phi i}^p \rightarrow \infty$$

so

$$k_{\phi i} \rightarrow k_{\phi i}^s$$

and

$$k_{\phi} \phi = Y(h - q^h)$$

$$\Delta Z_i t_i = Y_i q_i + k_{\phi i} \phi.$$

This is correct and is the most direct bridge to Chapter 9.

1.9 Chapter 3 to Chapter 9 bridge

We covered the main bridge correctly:

Chapter 3:

$$\Delta Z_i t_i = Y_i q_i + k_{\phi i} \phi$$

as a quasi-static roll/load-transfer closure.

Chapter 9:

$$\Delta Z_i t_i = Y_i q_i + k_{\phi i} \phi + c_{\phi i} \dot{\phi}$$

with roll becoming a true dynamic DOF.

Chapter 9 also adds:

$$\Delta Z = \frac{-ma_x h + J_y \ddot{\theta}}{l}$$

instead of Chapter 3's approximately

$$\Delta Z \approx -\frac{ma_x h}{l}.$$

And Chapter 9's total lateral load transfer becomes:

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = ma_y h + mg(h - d)\phi - J_x \ddot{\phi} + J_{xx} \dot{\phi}.$$

That relation was explained correctly.

2. What is still missing for “fully learning Chapter 3”

The previous answers are good for the load-transfer/no-roll-axis core, but Chapter 3 contains more than that. To fully learn Chapter 3, you still need these missing or under-explained parts.

2.1 Full congruence / kinematic equations

We explained a_x, a_y , but not enough of Chapter 3's kinematic framework:

$u, v, r \rightarrow \text{trajectory} \rightarrow \text{velocity center} \rightarrow \text{fundamental ratios} \rightarrow \text{acceleration center} \rightarrow \text{tire slips}.$

Chapter 3 spends substantial space on these before reaching load transfer. The contents show that Section 3.2 includes velocities, yaw angle and trajectory, velocity center, fundamental ratios, accelerations/radii of curvature, acceleration center, and tire kinematics. 978-94-017-8533-4

For learning, you should understand at least this:

β or $\hat{\beta}$ = vehicle slip angle

$$\rho = \frac{r}{u} \quad \text{or related curvature measure}$$

R, S = coordinates of the velocity center

These quantities are not just geometric curiosities. Later chapters use them to analyze handling, cornering curvature, MAP, and steady-state behavior.

2.2 Tire-slip formulas

We have not fully derived the four-wheel slip equations.

For example, the velocity at each wheel center depends on:

$$u, \quad v, \quad r, \quad a_x, \quad t_i, \quad \delta_{ij}.$$

The front-left wheel sees something like:

$$u - \frac{r t_1}{2}, \quad v + r a_1.$$

The front-right wheel sees:

$$u + \frac{r t_1}{2}, \quad v + r a_1.$$

The rear-left wheel sees:

$$u - \frac{r t_2}{2}, \quad v - r a_2.$$

The rear-right wheel sees:

$$u + \frac{r t_2}{2}, \quad v - r a_2.$$

Then these velocities are rotated into each wheel frame using δ_{ij} , producing longitudinal and lateral slips. The book later repeats the assembled model and shows the lateral slip formulas and roll-steer relation, including the dependence of steering angle on suspension roll angle. 978-94-017-8533-4

This part is necessary if you want to implement the Chapter 3 model.

2.3 Tire constitutive equations

We mentioned the generic tire law:

$$F_{xij} = F_{xij}(Z_{ij}, \gamma_{ij}, \sigma_{xij}, \sigma_{yij}, \varphi_{ij})$$

$$F_{yij} = F_{yij}(Z_{ij}, \gamma_{ij}, \sigma_{xij}, \sigma_{yij}, \varphi_{ij})$$

but we did not fully explain Chapter 3's tire-constitutive role.

The important idea is:

kinematics gives slips; tire model maps slips and loads to forces.

So the model loop is:

$$u, v, r, \delta, \omega, Z, \gamma \rightarrow \sigma_x, \sigma_y, \varphi \rightarrow F_x, F_y.$$

Without that, the load-transfer derivation is disconnected from actual vehicle dynamics.

2.4 Grip/friction limit

We did not fully explain the road-tire friction inequality.

The errata corrects the friction condition in Chapter 3 to:

$$X_{ij}^2 + Y_{ij}^2 \leq \mu_p Z_{ij}, \quad Z_{ij} \geq 0.$$

This is important because tire forces are not arbitrary. Vertical load Z_{ij} limits the possible combined longitudinal/lateral force. The errata explicitly corrects the Chapter 3 friction inequality. 978-94-017-8533-4

2.5 Suspension internal coordinates and camber variation

We discussed roll stiffness and tire vertical stiffness, but not enough of the suspension internal-coordinate derivation.

Chapter 3 introduces suspension internal coordinates because the suspension does more than create load transfer. It also changes:

γ_{ij} = camber angle,

δ_{ij} = steering angle, including roll steer,

z_i = vertical displacement,

Δt_i = track variation.

The book gives camber variation formulas and states an important warning:

different suspensions with the same no-roll center can behave differently.

2.6 Suspension jacking

We mentioned jacking, but did not derive it.

Jacking is the vertical motion/force effect associated with lateral forces passing through no-roll centers. The conceptual relation is:

lateral force through suspension links → vertical component/constraint effect → body lift or drop.

This is different from elastic roll. Jacking is one reason a force through the no-roll center can produce important vertical effects even though it produces no suspension-spring roll.

For your Chapter 9 understanding, jacking matters because Chapter 9 also uses no-roll/VIP ideas and explicitly criticizes the classical roll-axis interpretation.

2.7 Dependent suspensions

We did not explain Section 3.9. It is not central for your independent-suspension race-car model, but it is part of Chapter 3.

The main point is:

dependent suspensions have different kinematic constraints and different no-roll-center behavior.

If your vehicle model is independent suspension, you can treat this as secondary.

2.8 Sprung and unsprung masses

We mentioned Chapter 9 with unsprung mass, but we did not fully explain Chapter 3's Section 3.10.

Chapter 3 initially uses a simplified vehicle mass model, but then discusses sprung and unsprung masses. This matters because tire vertical loads are physically applied at the contact patch, while suspension forces act between sprung and unsprung parts.

For high-fidelity modeling, this distinction becomes essential:

$$m = m_s + m_u.$$

If unsprung masses are neglected, load-transfer equations are cleaner. If they are included, some vertical load components come from unsprung mass inertia and gravity directly.

For your 14DOF work, this is not optional. But for learning Chapter 3's first-order model, it can be postponed until after the main load-transfer derivation.

2.9 Differential mechanism

We did not explain Section 3.11.4.

Chapter 3 includes principles of differential mechanisms because longitudinal tire forces X_{ij} are affected by how drive/brake torques are distributed between left/right and front/rear wheels.

This matters because:

$$N_X = \Delta X_1 t_1 + \Delta X_2 t_2$$

appears in the yaw equation.

So if a locked or limited-slip differential creates left-right longitudinal force difference, it affects yaw moment and therefore changes:

$$N_Y = J_z \dot{r} - N_X$$

and therefore changes:

$$Y_1 = \frac{Y a_2 + N_Y}{l}, \quad Y_2 = \frac{Y a_1 - N_Y}{l}.$$

So the differential can indirectly change Y_1, Y_2 , then q^b , then lateral load transfer distribution.

We mentioned N_X , but not this full drivetrain connection.

2.10 Full model structure

We did say the model is a coupled loop, but it is worth stating again more explicitly.

The Chapter 3 model is not solved in a simple one-pass sequence. It is a coupled algebraic/dynamic system:

$$\begin{aligned} (u, v, r, \omega_{ij}, \delta_v) &\rightarrow \sigma_{xij}, \sigma_{yij}, \varphi_{ij} \\ (Z_{ij}, \gamma_{ij}, \sigma_{xij}, \sigma_{yij}, \varphi_{ij}) &\rightarrow F_{xij}, F_{yij} \\ F_{xij}, F_{yij} &\rightarrow X_i, Y_i, \Delta X_i \\ X_i, Y_i, \Delta X_i &\rightarrow \dot{u}, \dot{v}, \dot{r} \\ Y_i, N_Y, q_i, k_{\phi i} &\rightarrow \Delta Z_i \\ \Delta Z, \Delta Z_i &\rightarrow Z_{ij} \end{aligned}$$

then back into the tire model.

That closed loop is the actual Chapter 3 model.

3. The corrected answer to your question

So, have I provided all necessary information for you to learn Chapter 3?

For the lateral load-transfer/no-roll-axis core: yes.

For the entire Chapter 3 model: not yet.

What you already have is enough to understand the hardest conceptual part:

no-roll center → geometric load transfer → elastic roll moment → $\Delta Z_1, \Delta Z_2 \rightarrow Z_{ij}$.

But to fully learn Chapter 3, you still need a detailed derivation of:

wheel velocities → tire slips → tire forces → complete model closure.

You also need a compact but careful treatment of:

camber variation, roll steer, jacking, sprung/unsprung mass assumptions, differential-induced yaw moment.

4. Recommended learning order for you

Use this order, because it matches the logic of the chapter and prepares you for Chapter 9:

- Planar rigid-body kinematics
Learn $u, v, r, a_x = \dot{u} - vr, a_y = \dot{v} + ur$, wheel-center velocities.
- Tire slips
Derive $\sigma_{xij}, \sigma_{yij}, \varphi_{ij}$ from wheel velocities and steering angles.
- Tire force transformation
Convert F_{xij}, F_{yij} into X_{ij}, Y_{ij} .
- Global equilibrium
Derive Y_1, Y_2, N_X, N_Y , and the yaw equation.
- Longitudinal load transfer
Derive Z_1, Z_2 and ΔZ .
- Lateral load transfer global equation
Understand why $\Delta Z_1 t_1 + \Delta Z_2 t_2 = Y h$ is not enough.
- No-roll center/no-roll axis
Understand $Y q^b = Y_1 q_1 + Y_2 q_2$ and $Y h = Y q^b + Y (h - q^b)$.
- Suspension/tire internal equilibrium
Derive $\Delta Z_i t_i = Y_i q_i + k_{\phi i}^s \phi_i^s = k_{\phi i}^p \phi_i^p$.
- Individual tire loads
Compute $Z_{11}, Z_{12}, Z_{21}, Z_{22}$.
- Complete coupled model
Close the loop from tire loads to tire forces back to vehicle motion.

That sequence will make Chapter 9 much easier, because Chapter 9 mainly says:

now let roll and pitch be real dynamic coordinates instead of quasi-static internal load-transfer variables.